JUNIA Grande école d'ingénieurs

HEI·ISEN·ISA

Septembre -Décembre 2025

ISEN
Module
Numérique Responsable



Lesson 2

- Q/A MOOC if needed

- Information System
- Data Center & IA
- Standards

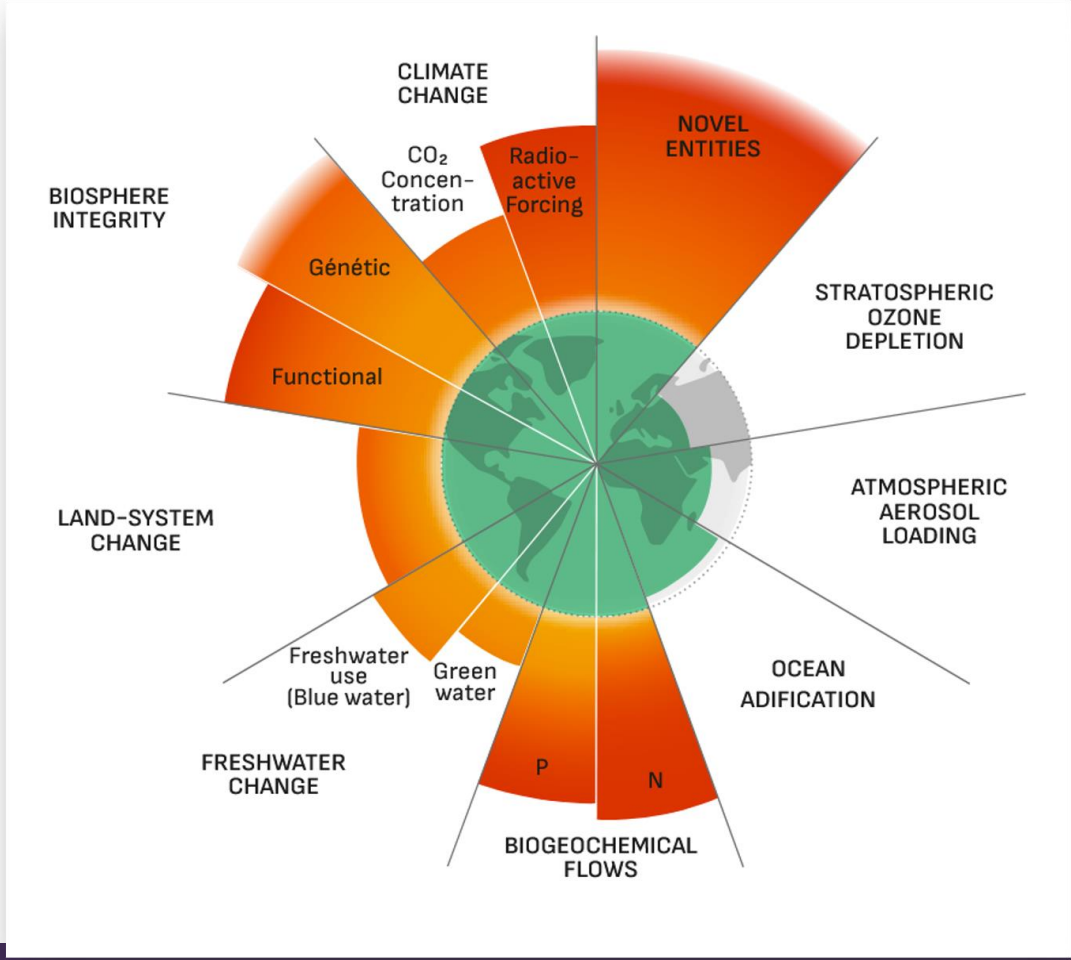
Just to test...



How many phones have been sold between 2000 and 2022 ?

31 billion !

Could you give at least 4 of the 9 planetary boundaries ?





For digital, what is the contribution of user equipments to global warming?

32 %

54 %

65 %



What percentage of WEEE end up in an illegal channel?

10 %

30 %

60 %

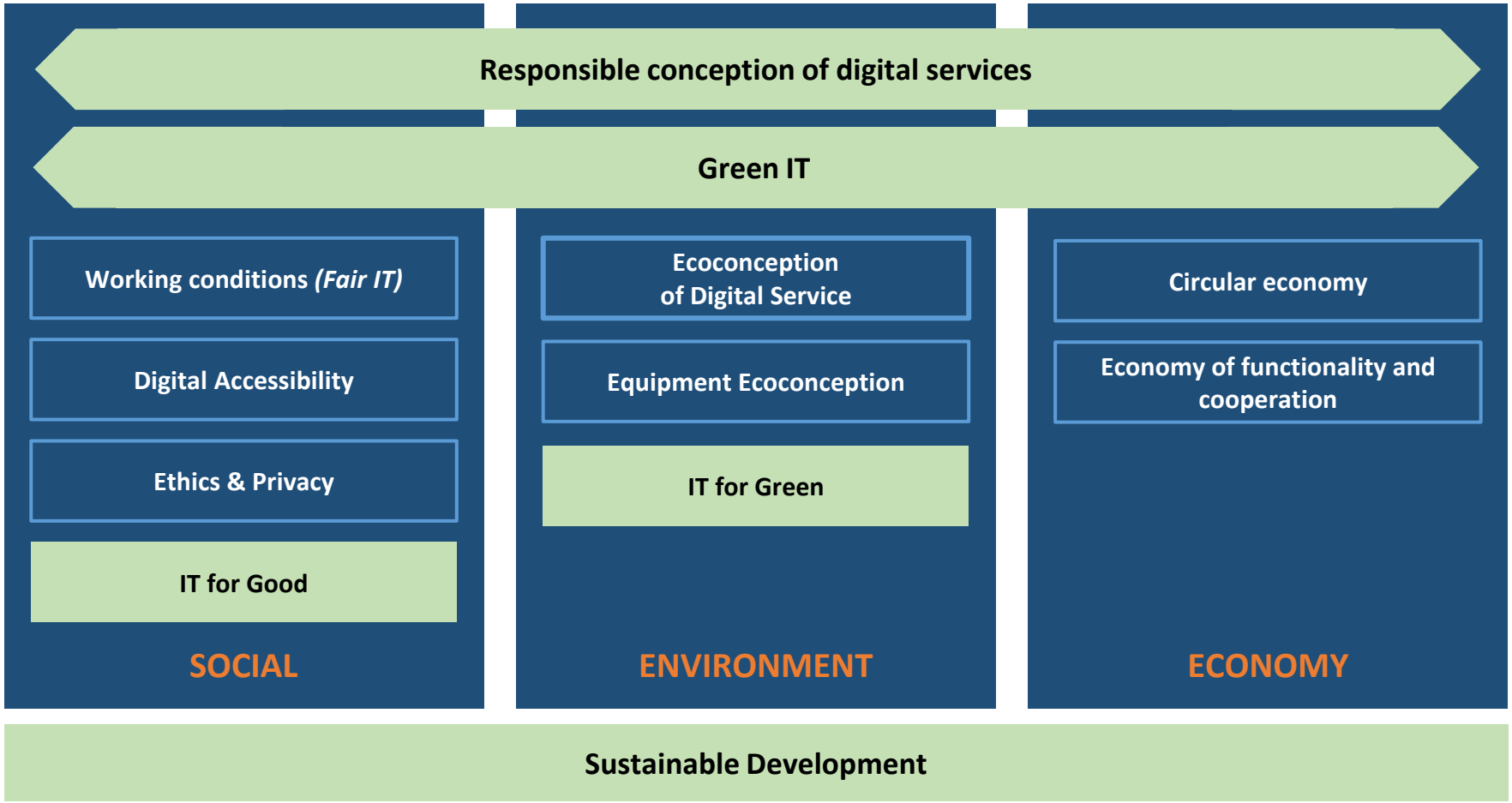


What part of global electricity does digital consume?

10 %

And the needs increase due to IA !

Responsible Digital



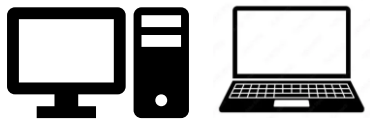
sources : Du Green IT au numérique responsable : lexique des termes de référence, collectif Green IT, 2018,

INFORMATION SYSTEM



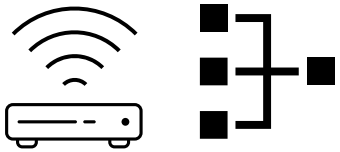
What are the different parts of an information system in a company ?

User environment





Telephony (landline, mobile)

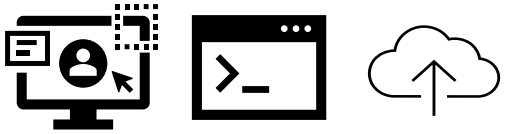


Network (LAN, WAN, WIFI)

Private data center



Cloud services
(IaaS, PaaS, SaaS)





Printing system

Some impacts figures…

1 laptop ~ 160 kg CO2eq
1 computeur high perf ~296 kg CO2eq
1 screen ~95 kg CO2eq

User environment



1 server ~ 600 kg CO2eq

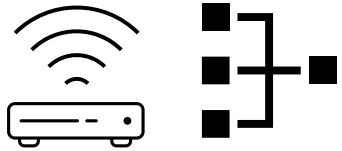
Private data center



1 smartphone
~ 80 kg CO2eq



Telephony (landline, mobile)

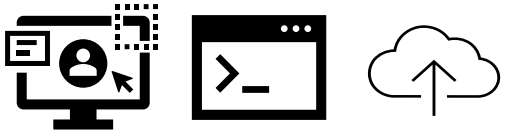


Network (LAN, WAN, WIFI)

1 switch ~ 81 kg CO2eq
1 router ~ 300 kg CO2eq*



Cloud services
(IaaS, PaaS, SaaS)





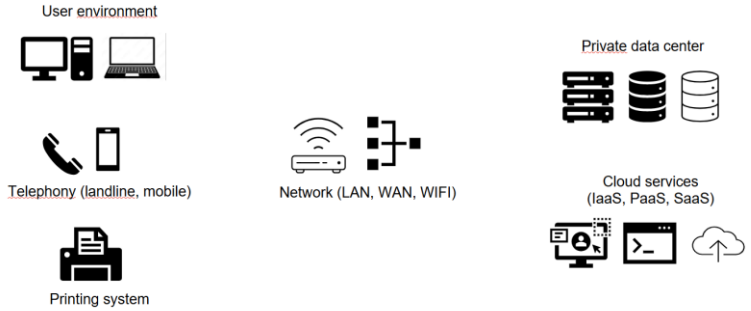
1 simple printer
~ 197 kg CO2eq

Printing system

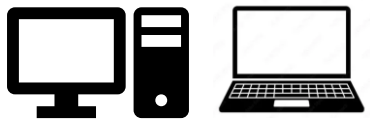
Source : https://base-empreinte.ademe.fr/donnees/jeu-donnees - *Core switch : Cisco reference – 1700 kg CO2eq for 5 years use



Which teams manage the different parts of the information system ?



User environment

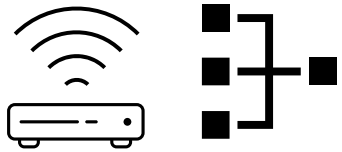




Telephony (landline, mobile)

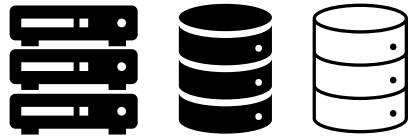


Printing system

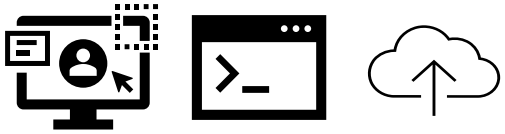


Network (LAN, WAN, WIFI)

Private data center



Cloud services
(IaaS, PaaS, SaaS)



Product Team | Business Team | General services | Outsourcing | HelpDesk | Network | ...

And not only one system....

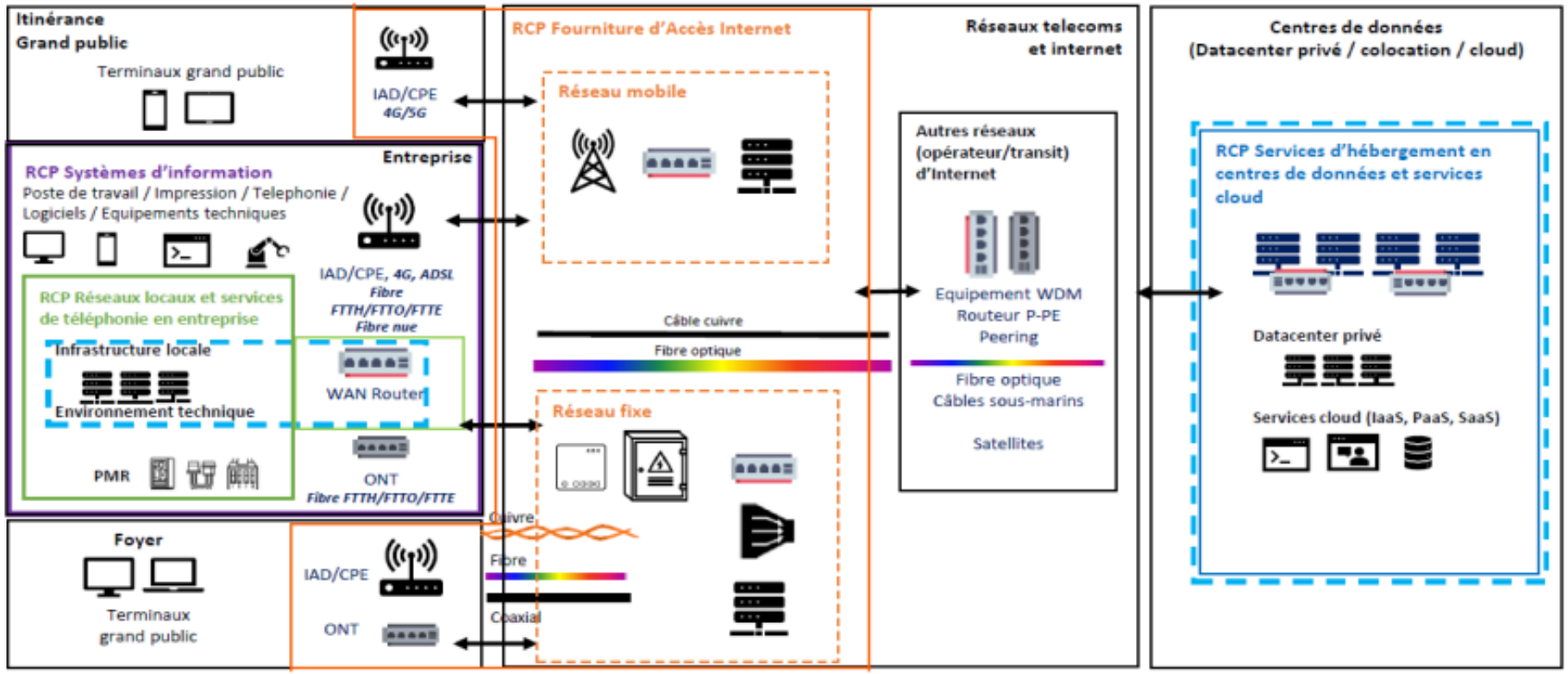


Figure 4: Schéma fonctionnel de l'articulation entre les différents RCP d'un service numérique

Légendes :

Bleu clair : Centres de données et services cloud – tiers couvert par le présent référentiel

↔ Echange de données

How to design a sustainable information system ?

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How to design a sustainable digital strategy....

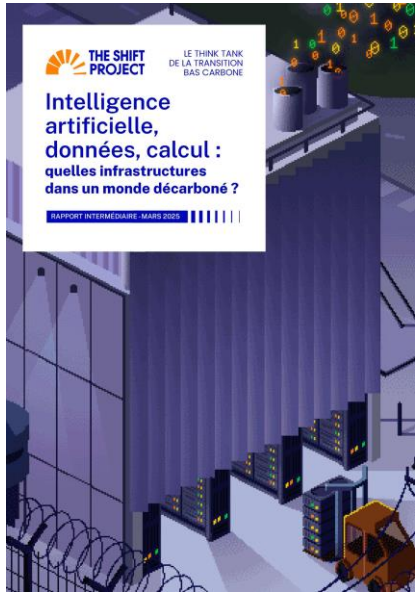


- Company culture
- Company commitment
- Many teams
- Different strategy
- Employee behaviour
- Stakeholders, providers
- Tools, Time, Money...

- Reduce environmental impact
- Reduce social impact
- Maintain the business

A lot of inspiration : analysis and good practices

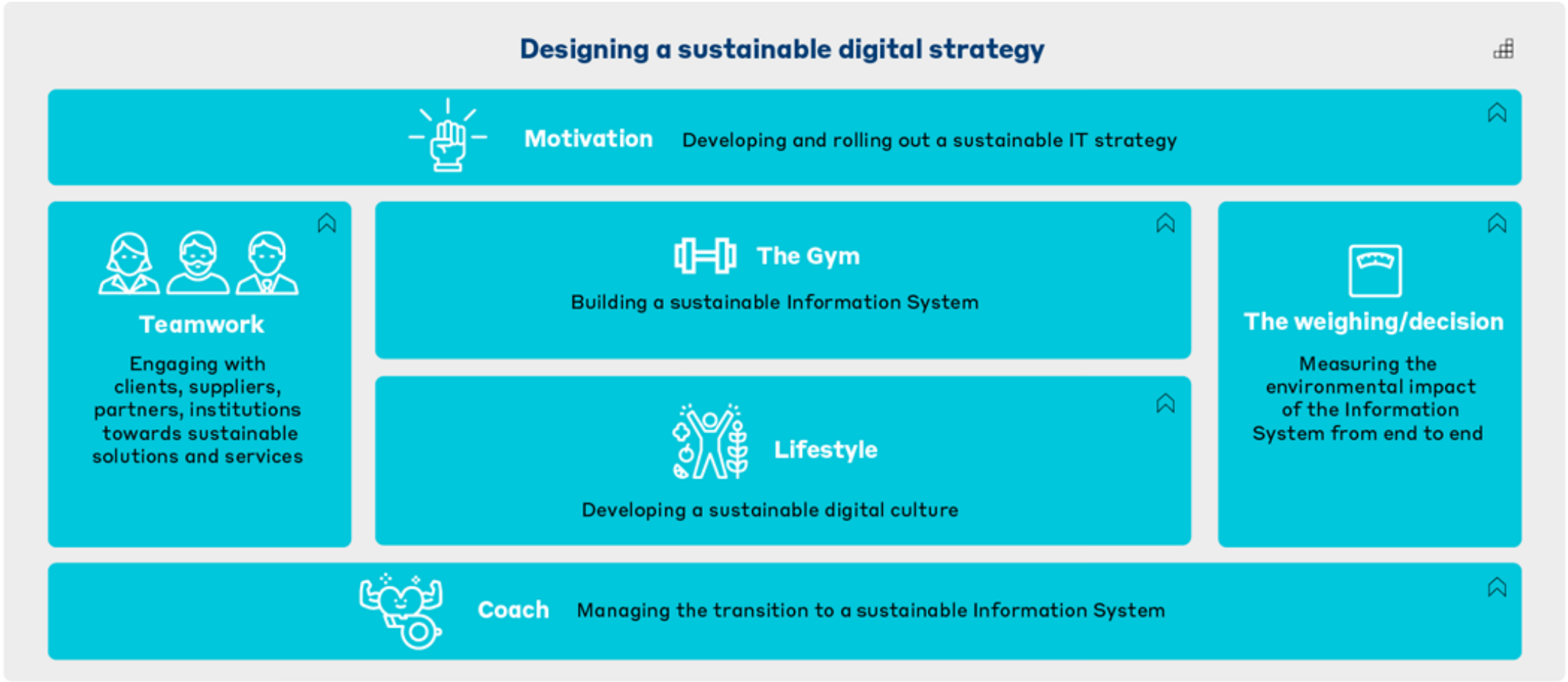






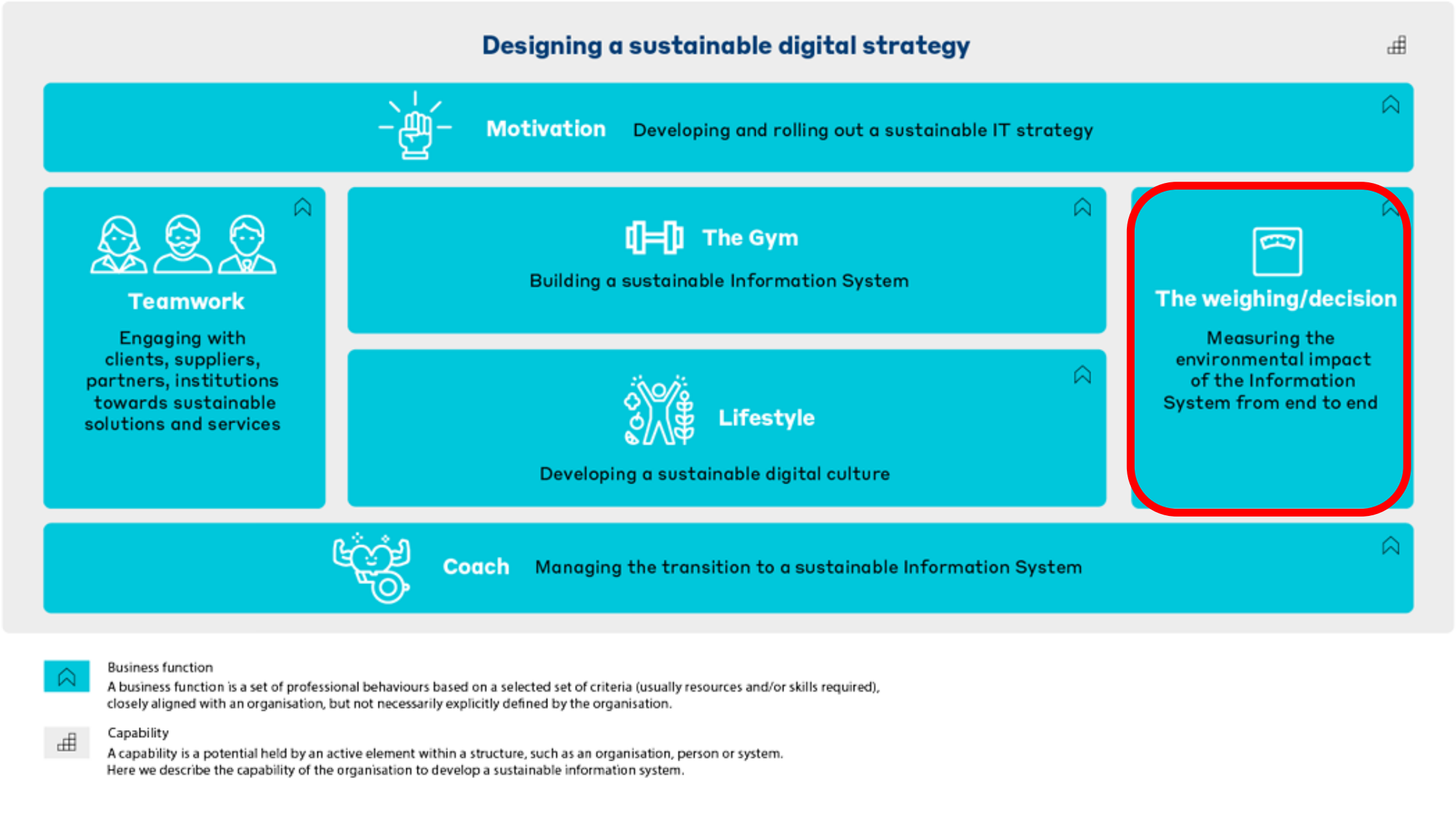


Sources : https://theshiftproject.org/publications/?t=numerique - https://www.cigref.fr/publications



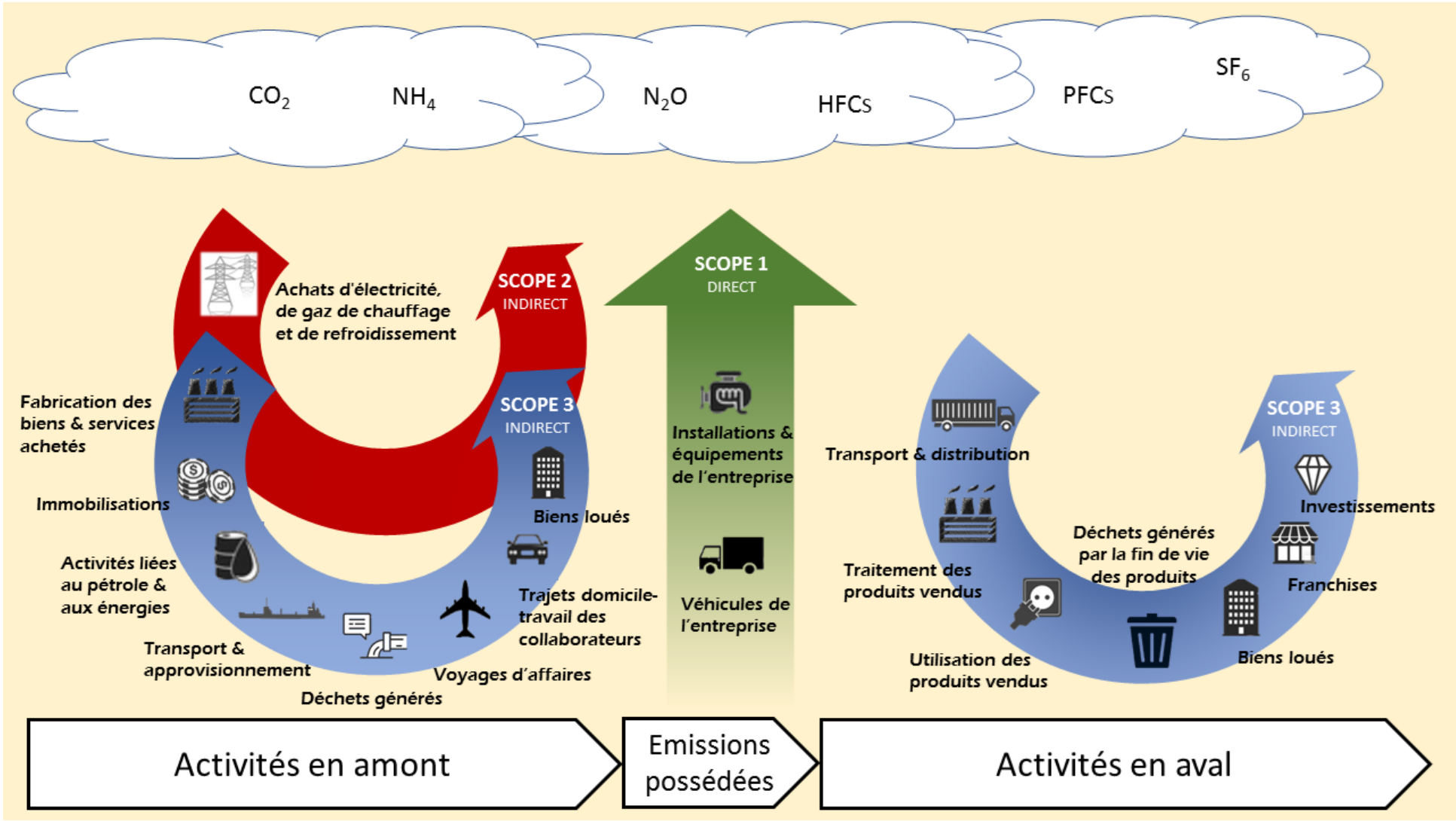
Reference model - high level view - “Get the Information System back in Shape”. [Source: The Shift Project, production of the working group]

MEASURE



Reference model - high level view - "Get the Information System back in Shape". [Source: The Shift Project, production of the working group]

Carbon Footprint of a company



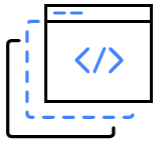
IS : What to measure?

Infrastructure Measure



- Terminals
- Network Equipment
- Datacenter

Services/Applications Measure



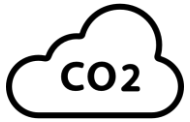


- Webconference
- To send a mail
- Web Request
- Product Publish
- ….

What to measure

Mono-criteria

CO2 emission



Multi criteria

Water consumption



Use of ressources



Electric consumption*



Consumption of primary energy

Main impact criteria monitored in digital technology

*Electricity consumption is not an environmental indicator; it is the consumption of fossil or renewable resources used to produce energy that is.

How to measure ?

Infrastructure Measurement

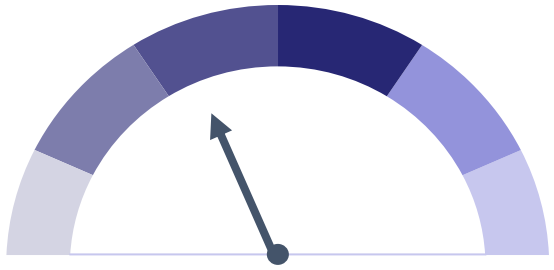
Macro Measurement

Theoretical Measurement

Single-Criteria Measurement

Manual Measurement

Expert Measurement



Service/Application Measurement

Detailed Measurement

Physical Measurement

Multi-Criteria Measurement

Automated Measurement

Autonomous Measurement

What is the priority?

Where should we place the cursor?

How to measure ?

Macro Measurement / Theoretical

Impact Data Details - ADEME Impact Database*

Terminal Equipment
- Laptop
- Tablet
- Smartphone
- Desktop Computer (without screen)
- Computer Monitor

Network Equipment
- 1 GB Transfer on Fixed Network
- 1 GB Transfer on Mobile Network

Data Center Equipment
- Server
- Virtual Machine - Small
- Virtual Machine - Medium
- Virtual Machine - Large

Detailed Measurement / Physical Measurement

- Laptop Dell Latitude E7285
  282 kgCO2e +/- 49 kgCO2e – 4 years of use
- Routers Cisco serie 800
  1700 kg CO2e - 5 years of use
- Server DELL PowerEdge C6420
  6510 kgCO2e - 4 years of use
- Company context > Factual usage duration.

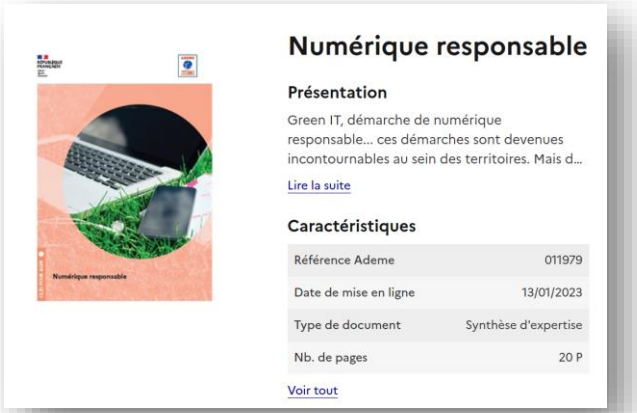
*Info : https://ecoresponsable.numerique.gouv.fr/posts/publications-donnees-negaoctet/

Where to find data ?

Data Reference

ADEME
Agence **D**e l'**E**nvironnement
et de la **M**aîtrise de l'**É**nergie

- Studies and publishing guides for sustainable digital (and many other domains)





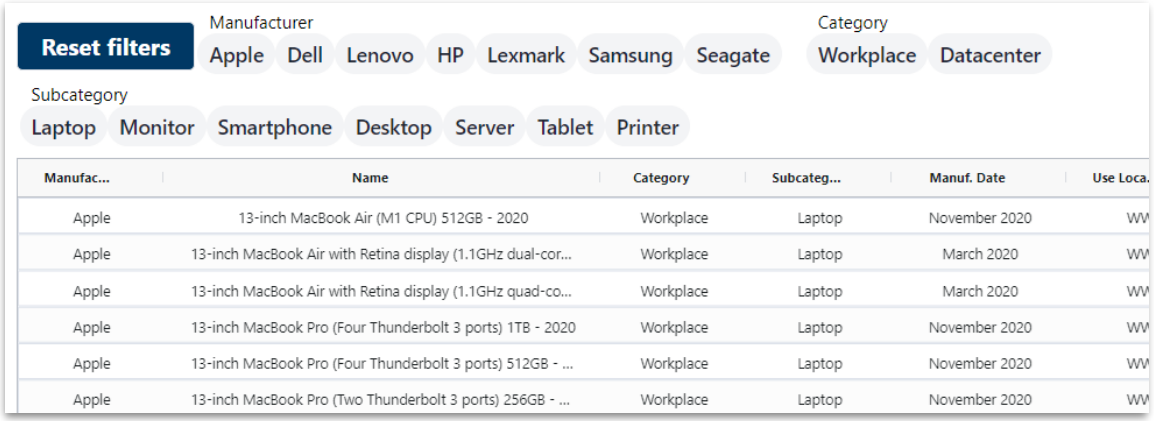
- Datasets
- Carbon database administrator

BASE Empreinte®

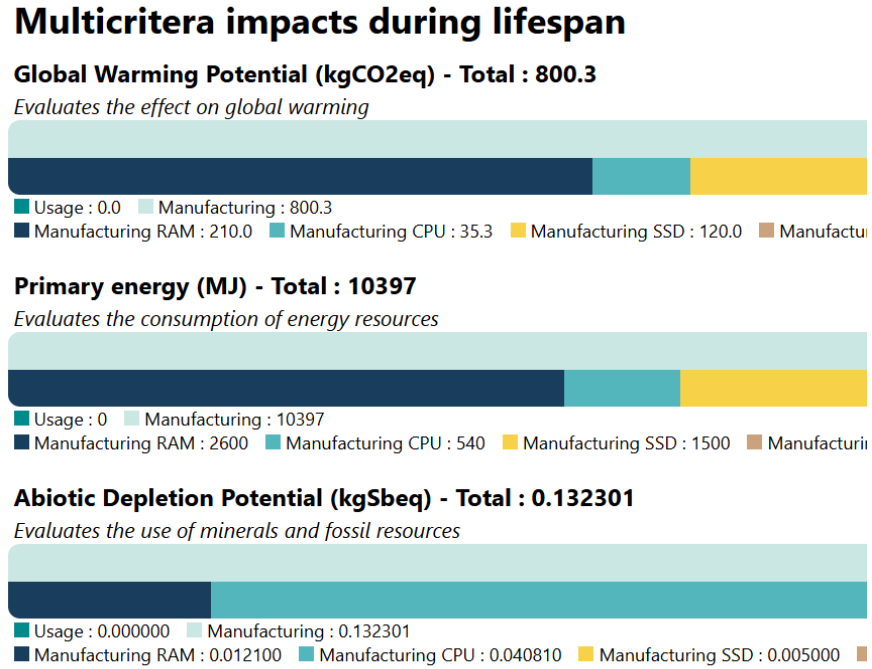
Data reference

Boavizta

Manufacturer data
https://dataviz.boavizta.org/



Assessment according to each configuration
https://dataviz.boavizta.org/serversimpact



https://github.com/Boavizta/environmental-footprint-data

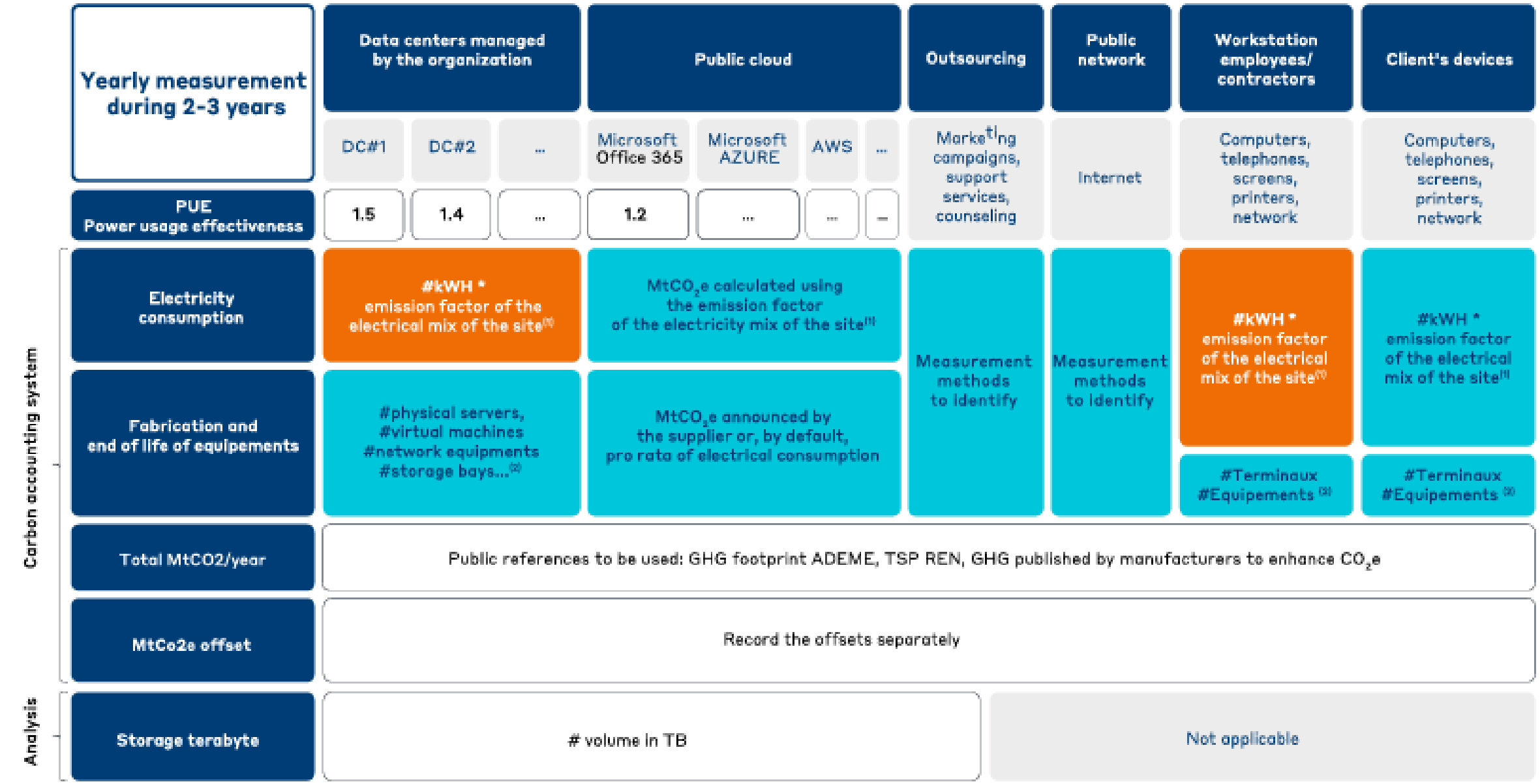
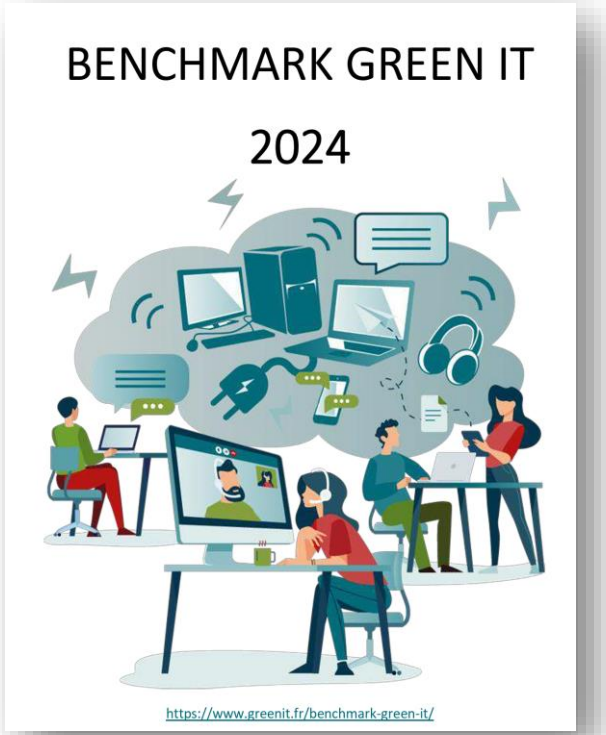


Figure 16 - Yearly environmental assessment of the information system - model inspired by the work carried out by the AXA group in 2020 as part of the "Digital Sustainability" program to measure and manage the impact of IT

Source: The Shift Project, production of the working group

To compare & improve

A French study about environmental impacts of organizations' information systems



Benchmark Green IT, since 2016

- 8 editions
- 65 participating organisations
- 4 countries

👉 **https://www.greenit.fr/benchmark-green-it/**

Green IT benchmark Perimeter

Functional Unit

"Make available and use the information system of organization** XXX**, by all of its users, for one year."

Information System '3 Tiers' :

- User environment
- Network
- Data Center

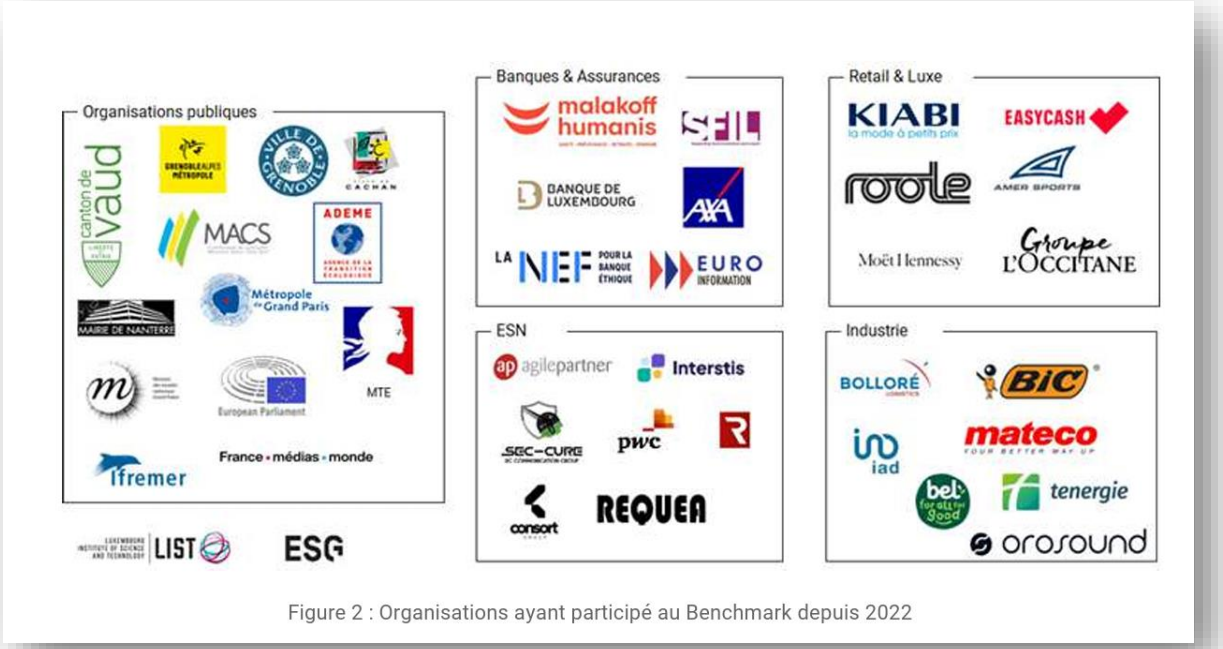


Figure 2 : Organisations ayant participé au Benchmark depuis 2022

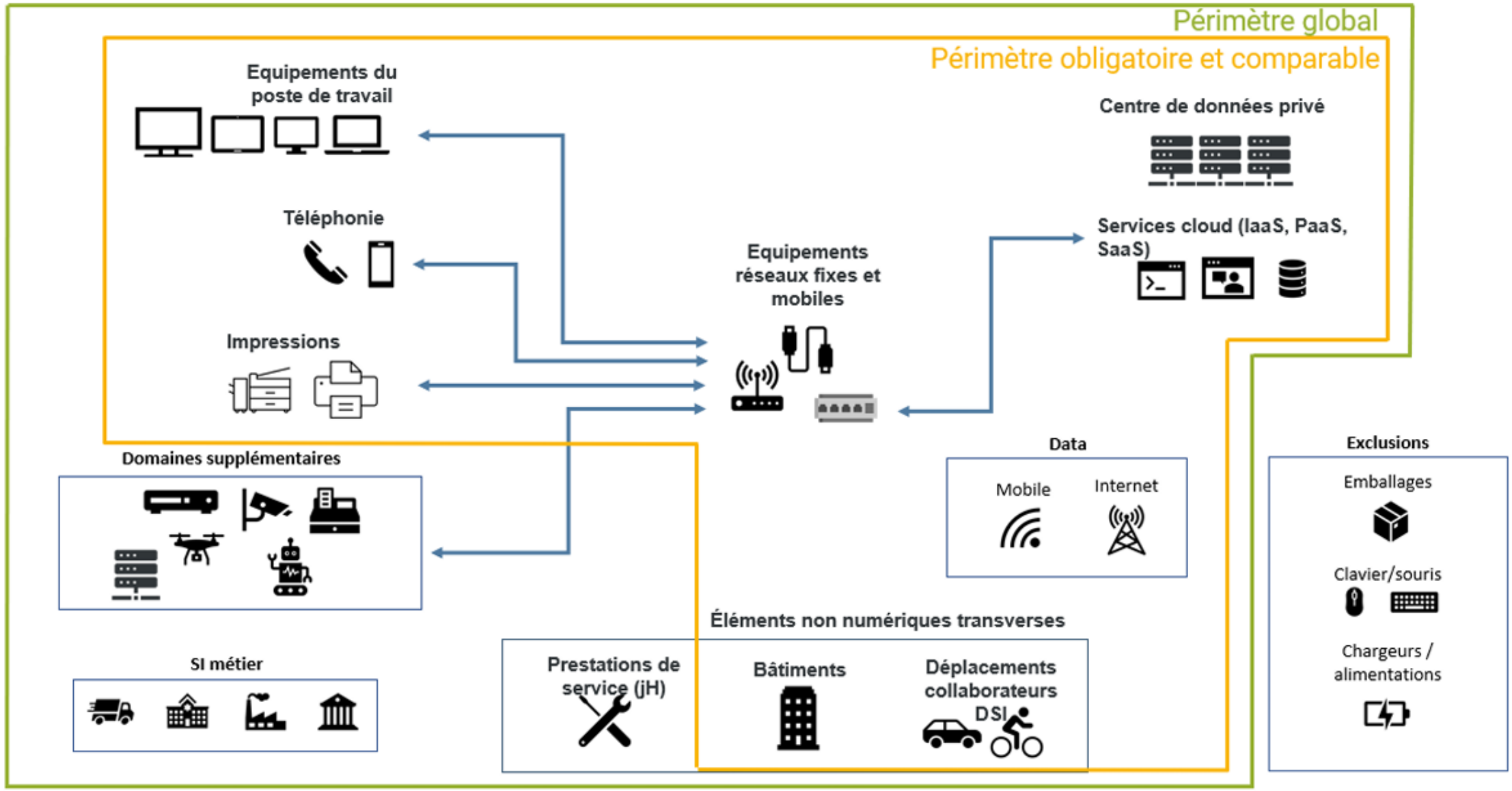


Figure 1 : Schéma du périmètre fonctionnel

Green IT benchmark Perimeter

The inventory reflects the operation of the information systems of the 30 organizations from 2022 to 2024:

- 30 Organizations
- **154 537 Users**
- 12 398 IT employees (including service providers)
- 125 683 m² of offices dedicated to the IT Department
- **755 601 digital devices**

Green IT benchmark Perimeter

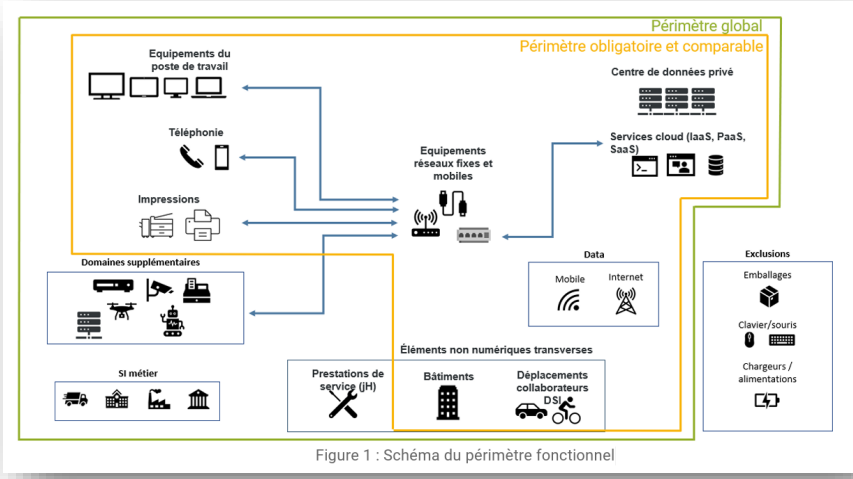


Figure 1 : Schéma du périmètre fonctionnel

INCLUSION

In order to be able to compare companies with each other and over time, we have chosen to establish a partial footprint, essentially corresponding to their management information system. Participating organizations are, however, invited to work on their total footprint, which represents their entire information system.

The following equipment and flows are taken into consideration:

- Non-digital cross-functional elements (DSI): Employee travel and office space;
- User work environment: smartphones, desktop and laptop computers, screens, etc. with the exception of video projectors;
- Printing: shared and personal printers, paper;
- Local network: IT equipment linked to the local area network (LAN);
- Wide area network (WAN) and mobile network (2G/3G/4G/5G);
- Cloud: VMs, storage
- Data centers: computing servers, storage arrays, network equipment, etc.

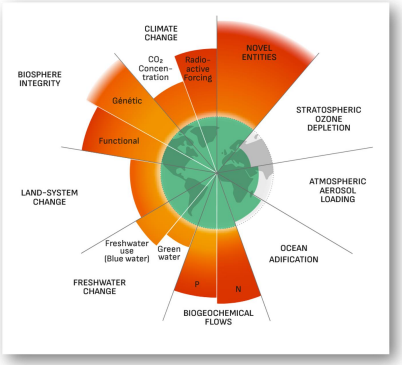
EXCLUSION

The following are excluded from the environmental assessment:

- Flows related to R&D and industrial IT
- Hosting services offered to third parties by the organization
- Equipment packaging and its end-of-life (remember, terminals have their own sector-specific rules), excluding packaging of equipment made available to users and its end-of-life;
- Broken or defective equipment;
- Construction and maintenance of infrastructure (building);
- Lighting, heating, sanitation, and cleaning of infrastructure (including IT);
- Transportation systems and infrastructure other than those dedicated to IT;
- Equipment installation.
- Purchases of services were calculated for each organization but do not fall within the comparable scope.

All of the above elements are considered to be outside the scope of the study.

Indicateurs environnementaux



Epuisement des ressources naturelles (minérales et métaux)
- Abréviation : PEF-ADPe
- Unité : kg équivalent Sb (kgeqSb)

Cet indicateur évalue la quantité de ressources (minéraux et métaux) retirées de la nature comme si elles étaient de l'antimoine.

Utilisation de la ressource en eau
- Abréviation : PEF-WU
- Unité : m3 world eq

Impact lié à la consommation d'eau douce

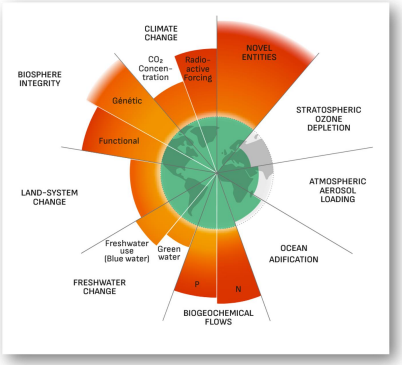
Changement climatique
- Abréviation : GWP
- Unité : kg équivalent C02 (kgeqCO2)

Les gaz à effet de serre (GES) sont des composés gazeux qui absorbent le rayonnement infrarouge émis par la surface de la Terre. L'augmentation de leur concentration dans l'atmosphère terrestre contribue au réchauffement global de la planète.

Emission de particules fines
- Abréviation : PEF-PM
- Unité : Disease incidence

La présence dans l'air de particules fines de petit diamètre, en particulier celles dont le diamètre est inférieur à 10 microns - représente un problème de santé humaine, car leur inhalation peut entraîner des problèmes respiratoires et cardiovasculaires.

Indicateurs environnementaux



Acidification (lacs, rivières ou eaux souterraines);
- Abréviation : PEF-AP
- Unité : mol H+ eq

L'acidification de l'air est liée aux émissions d'oxydes d'azote, d'oxydes de soufre, d'ammoniac et d'acide chlorhydrique.
Ces polluants se transforment en acides en présence d'humidité, et leurs retombées peuvent endommager les écosystèmes ainsi que les bâtiments.

Radiations ionisantes
- Abréviation : PEF-IR
- Unité : kBq U235 eq

Les radionucléides peuvent être libérés lors d'un certain nombre d'activités humaines.Lorsque les radionucléides se désintègrent, ils libèrent des rayonnements ionisants.
L'exposition humaine aux rayonnements ionisants endommage l'ADN, ce qui peut entraîner divers types de cancer et de malformations congénitales.

Epuisement des ressources abiotiques (fossiles)
- Abréviation : PEF-ADPf
- Unité : MJ

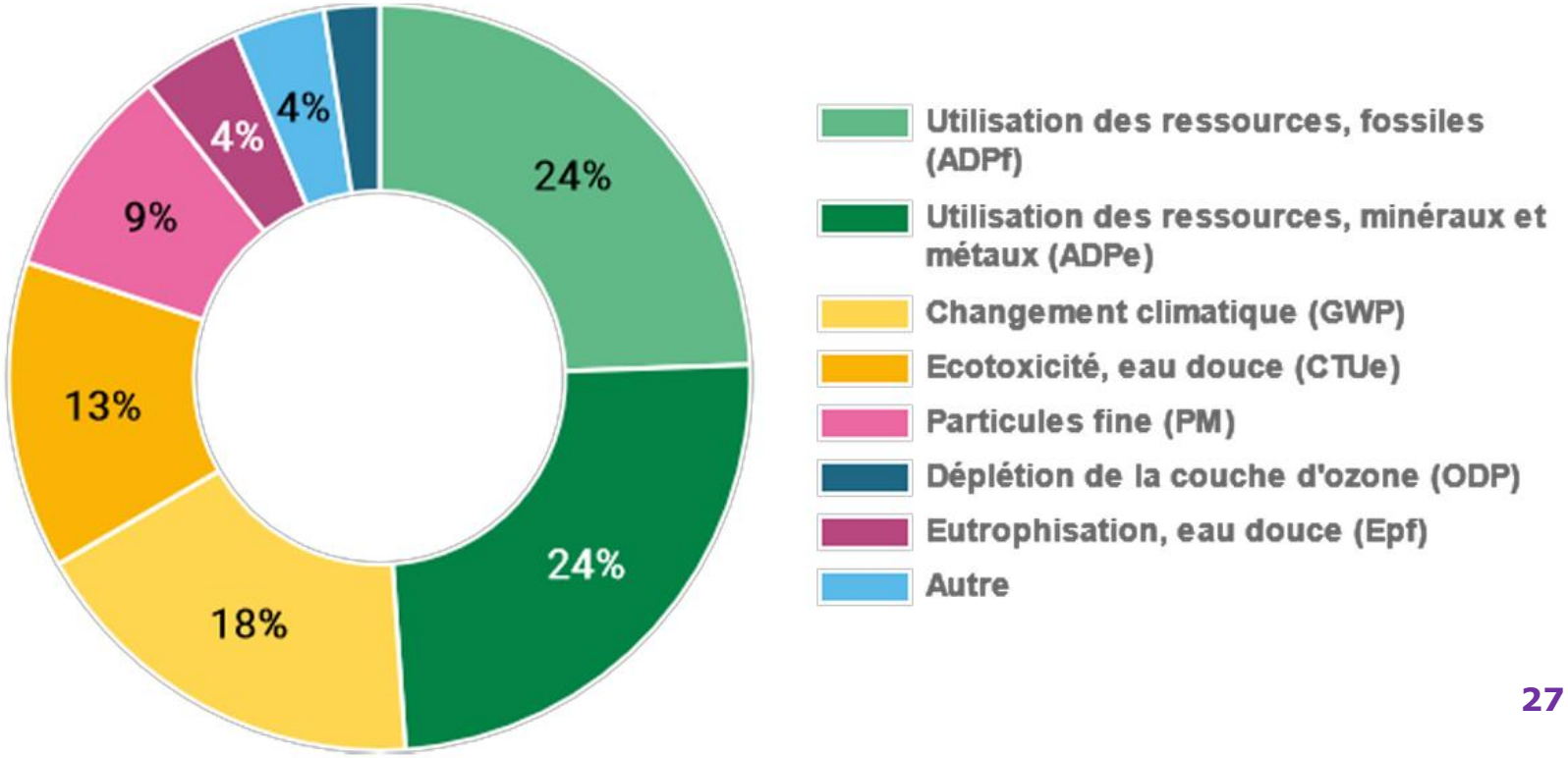
L'indicateur représente la consommation d'énergie primaire provenant de différentes sources non renouvelables (pétrole, gaz naturel, etc.).

Consommation d'énergie primaire
- Abréviation : CED
- Unité : MJ énergie primaire cumulée.

L'énergie primaire est la première forme d'énergie directement disponible dans la nature avant toute transformation : bois, charbon, gaz naturel, pétrole, vent, rayonnement solaire, énergie hydraulique ou géothermique, etc.

Green IT benchmark Results

Indicator contribution to the footprint : ***"Make available and use the information system of organization** XXX**, by all of its users, for one year."***

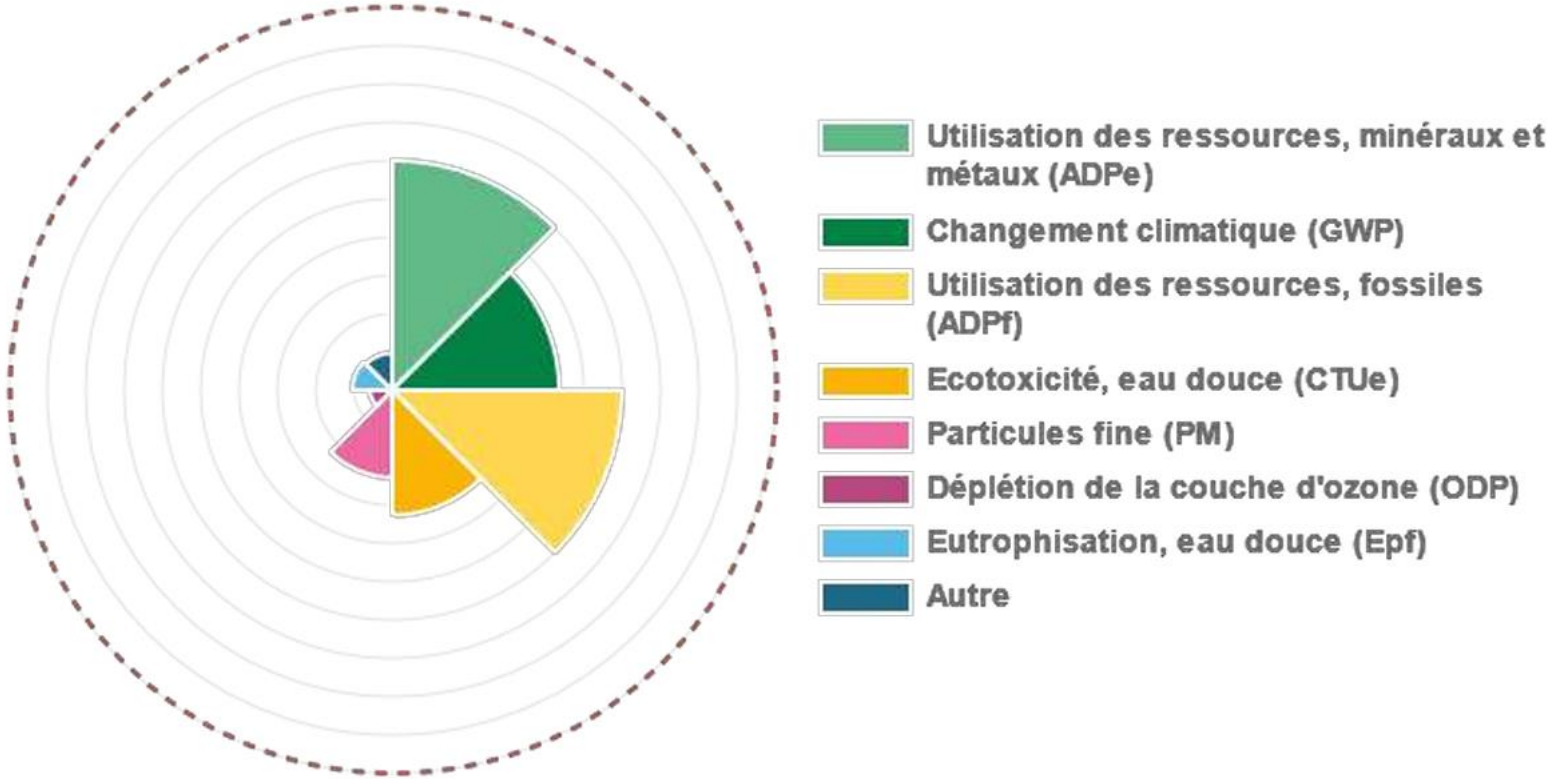


7,3 / year / user

27 showers / day / user

Green IT benchmark Results

Indicator contribution according to planetary boundaries



Distribution of impacts by domain and indicators

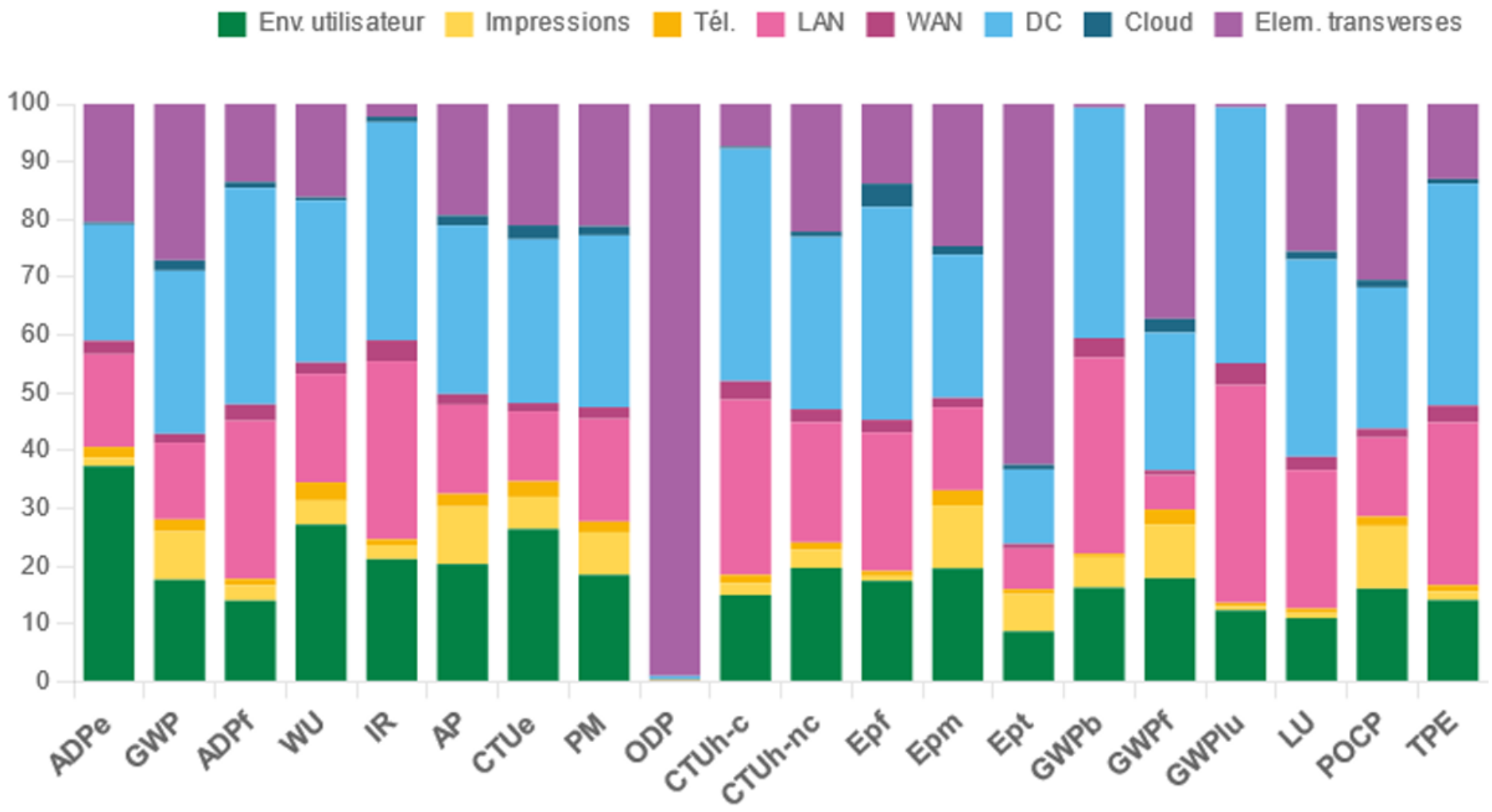
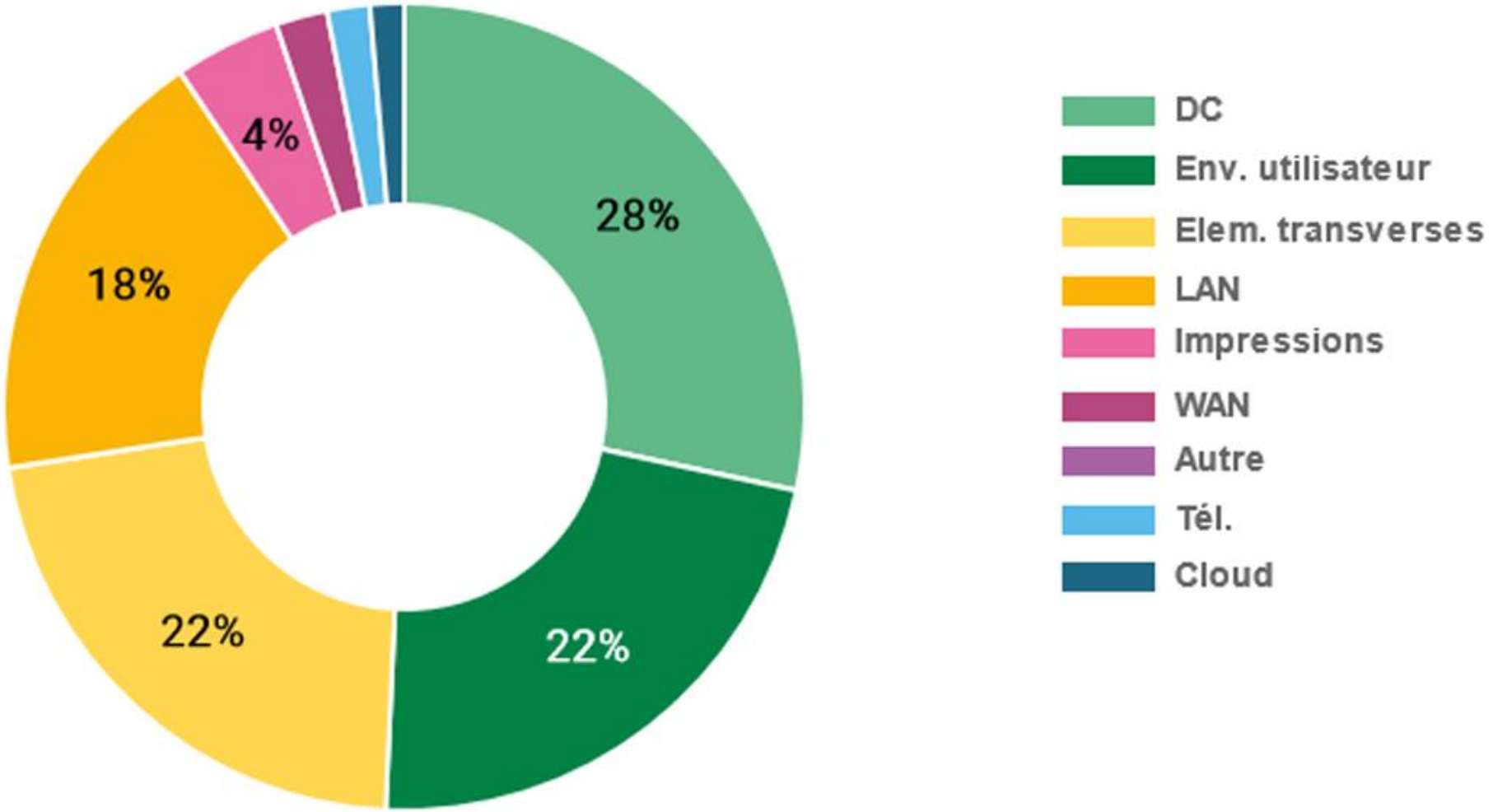
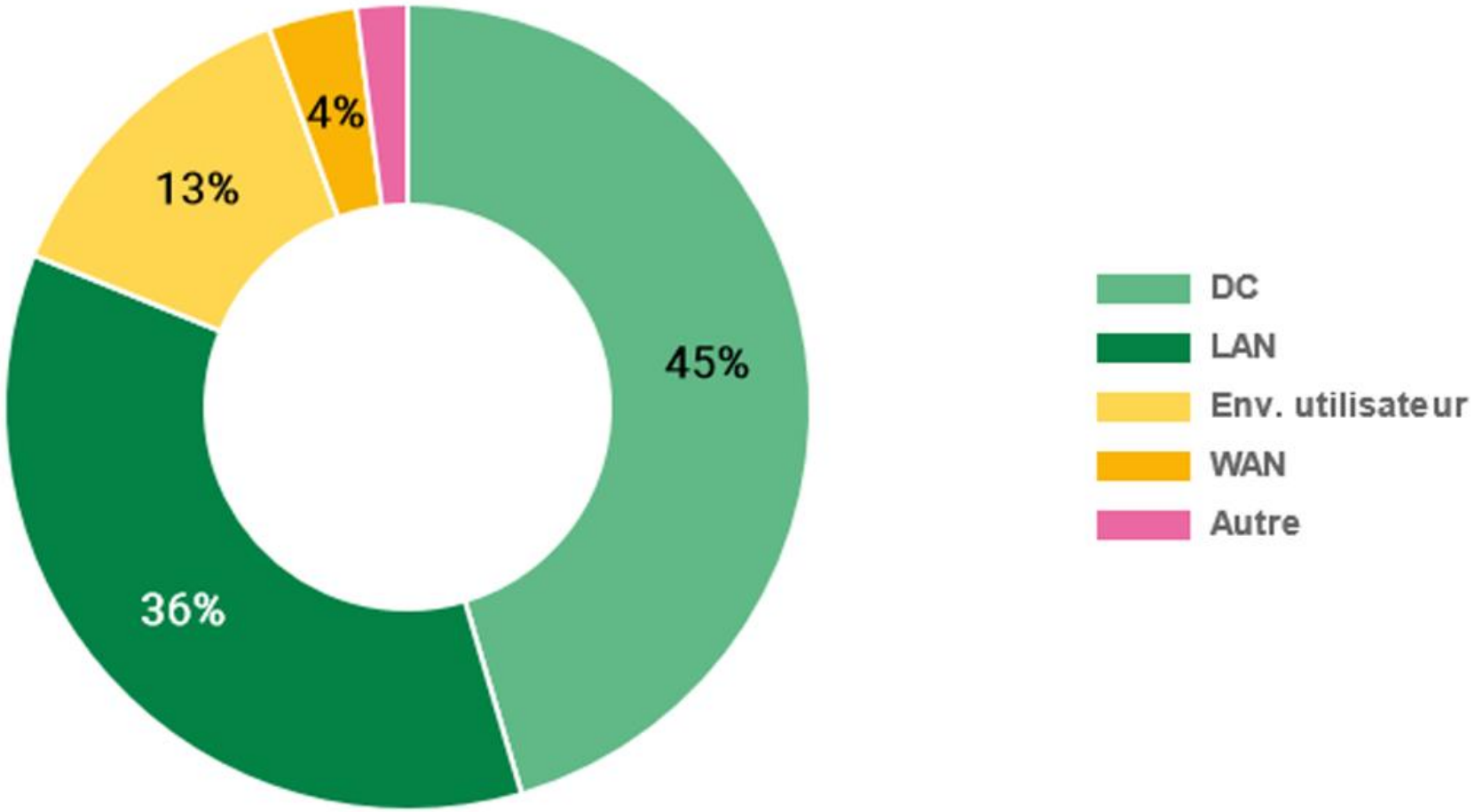


Figure 4 : Représentation des résultats par indicateur et par domaine du SI

Distribution of global footprint by domain



Electric consumption



Observations



- Stabilization of the **number of equipment / user** ~3,2

- Landline fall drastically > softphone

Smartphone deployment is not so high and not equivalent to landline furniture.

- **Printing** restart to increase : 10 pages / user / day

- **Network** : constant increase of number of equipments (WIFI deployment and better tracking)

- **Datacenter** : On premise PUE ~1,77

- **Storage** : 2 To / user

Improvement : Good practices



- **Continue to reduce the equipment rate and extend its lifespan;**
- **Reduce infrastructure electricity consumption (81% of total expenses)**
- Telephony and printing are areas with low environmental impacts (compared to other areas) but with easy actions to deploy.
- Reduce the equipment rate per user, **particularly by avoiding the use of second screens**
- Extend the lifespan of all equipment (including infrastructure) by systematically **reusing rather than recycling** functional equipment;
- Reduce the kilometers traveled by IT department and service provider employees by promoting remote working and implementing a Mobility Plan (PDM);
- Eco-design digital services to reduce the impacts associated with use by reducing the number of servers and the amount of "networks" required.

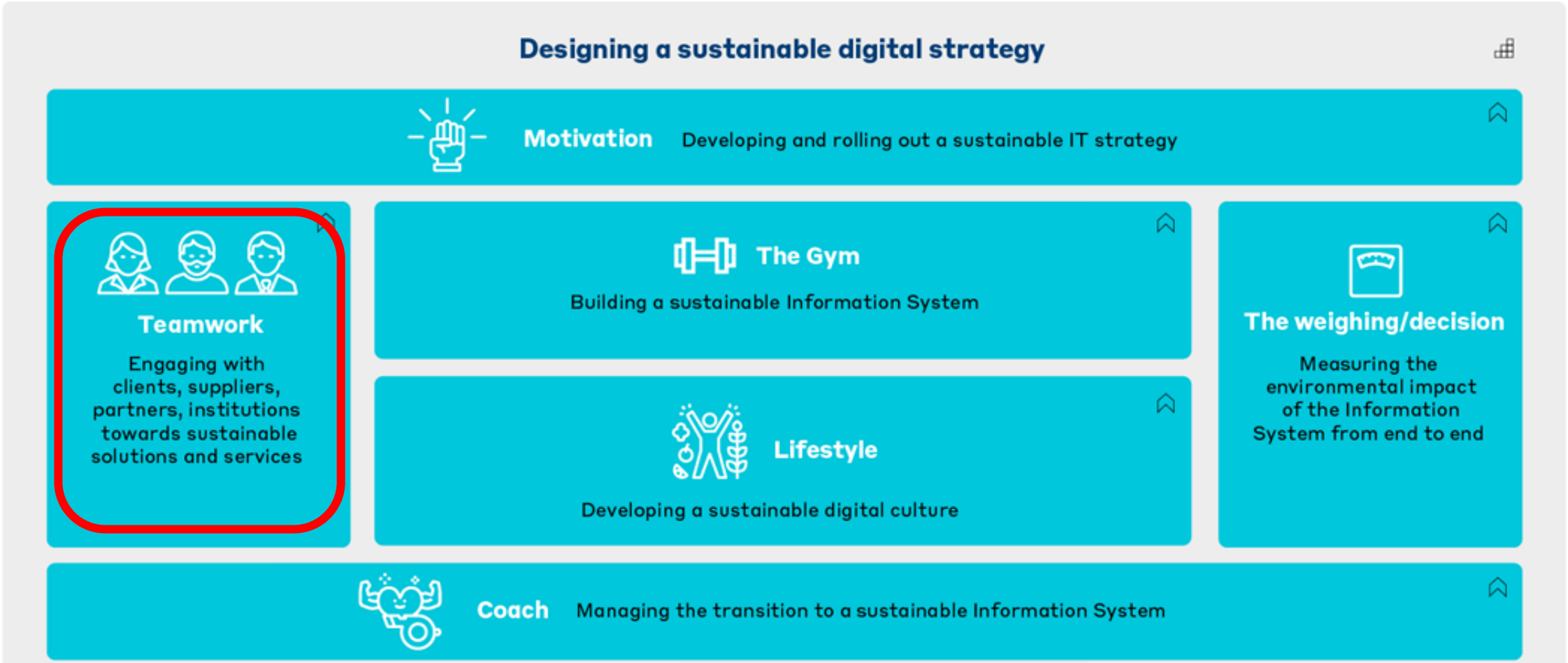


And your digital impact as Junia student ?

https://myimpact.isit-europe.org/fr

https://agirpourlatransition.ademe.fr/particuliers/bureau/numerique/calculez-lempreinte-carbone-usages-numeriques

PURCHASE



Reference model - high level view - "Get the Information System back in Shape". [Source: The Shift Project, production of the working group]

Indirect Purchase

RÉPUBLIQUE FRANÇAISE
Liberté
Égalité
Fraternité

mi num_ eco

DesignGouv
Par le pôle design des services numériques

- Regulations
- Digital Accessibility Framework
- Practical Guides for Organizations



Towards digital sustainable purchases



Selecting suppliers is a key step

Evaluate the maturity level of a supplier's CSR approach:

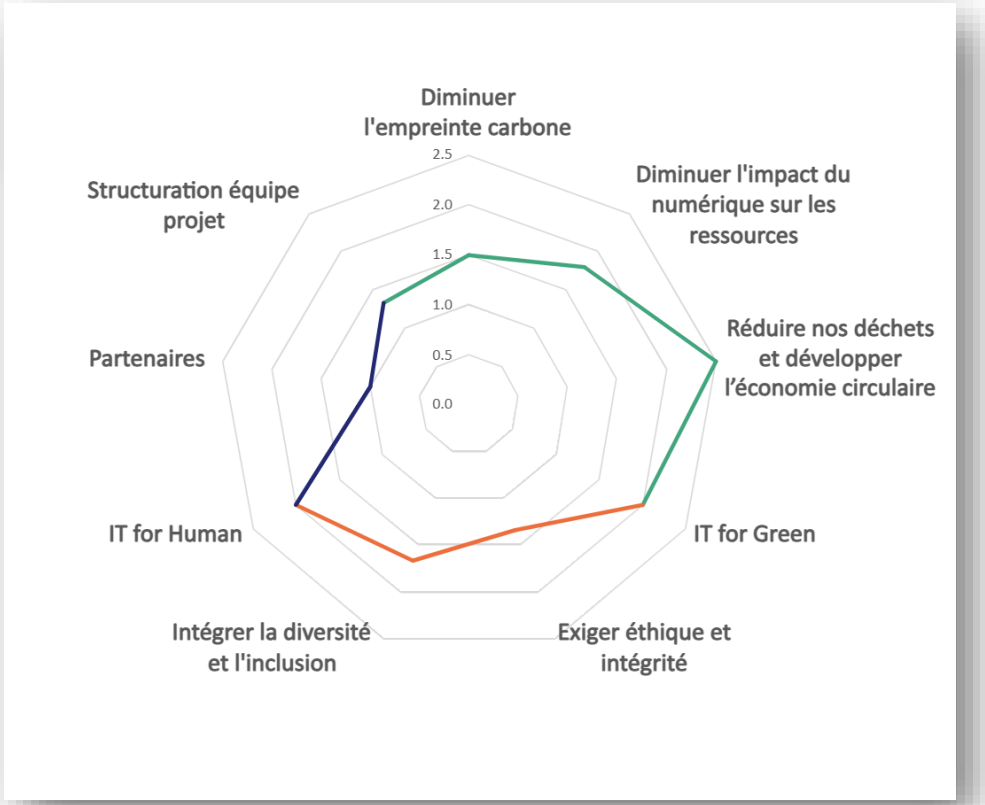
- **Governance:** strategy, organization, and communication within the company and its stakeholders.
- **Environment:** How is integration of environmental performance into suppliers' activities (measurements, impact reduction, process optimization, waste management, product eco-design, etc.).
- **Fair practices:** Recommendations relating to the fight against corruption and unfair competition.
- **Human Rights:** Recommendation on implementing measures to ensure respect for human rights throughout the value chain, as well as measures to prevent discrimination and promote diversity and gender equality.
- **Local Development:** Recommendation on implementing actions to promote regional development.

Challenge a project with CSR criteria

Diminuer l'empreinte carbone	Le projet est compatible avec l'atteinte des objectifs de la Stratégie Bas Carbone de l'entreprise (baisse de X% des émissions en 20XX)
	Le projet est toujours rentable si un prix du CO2 allant de 50 à 80€ la tonn est intégré au business plan (ou au budget alloué)

Example of a tool for evaluating a new project based on its environmental, social and governance impact

Outil-devaluation-RSE-et-projet-IT_VF-1.xlsx



Source : https://www.cigref.fr/criteres-de-decisions-rse-a-integrer-dans-les-projets-it-outil-devaluation-disponible

Towards digital sustainable purchases

Sizing and defining specifications are a key step, also from an environmental perspective.
Carrying out a functional analysis of needs and "lean" sizing represents a significant lever for improving environmental performance.

$$R = \frac{\text{(Acquisition cost) + (Usage costs * accounting life) +(Destruction cost)}}{\text{Accounting Life}}$$

The lower is R index, the better the product has a **cost/life cycle ratio.**
The purpose of this data is to allow a comparative analysis of supplier offers, thus facilitating the choice of the optimal equipment/services.

Towards digital sustainable purchases eco-labels

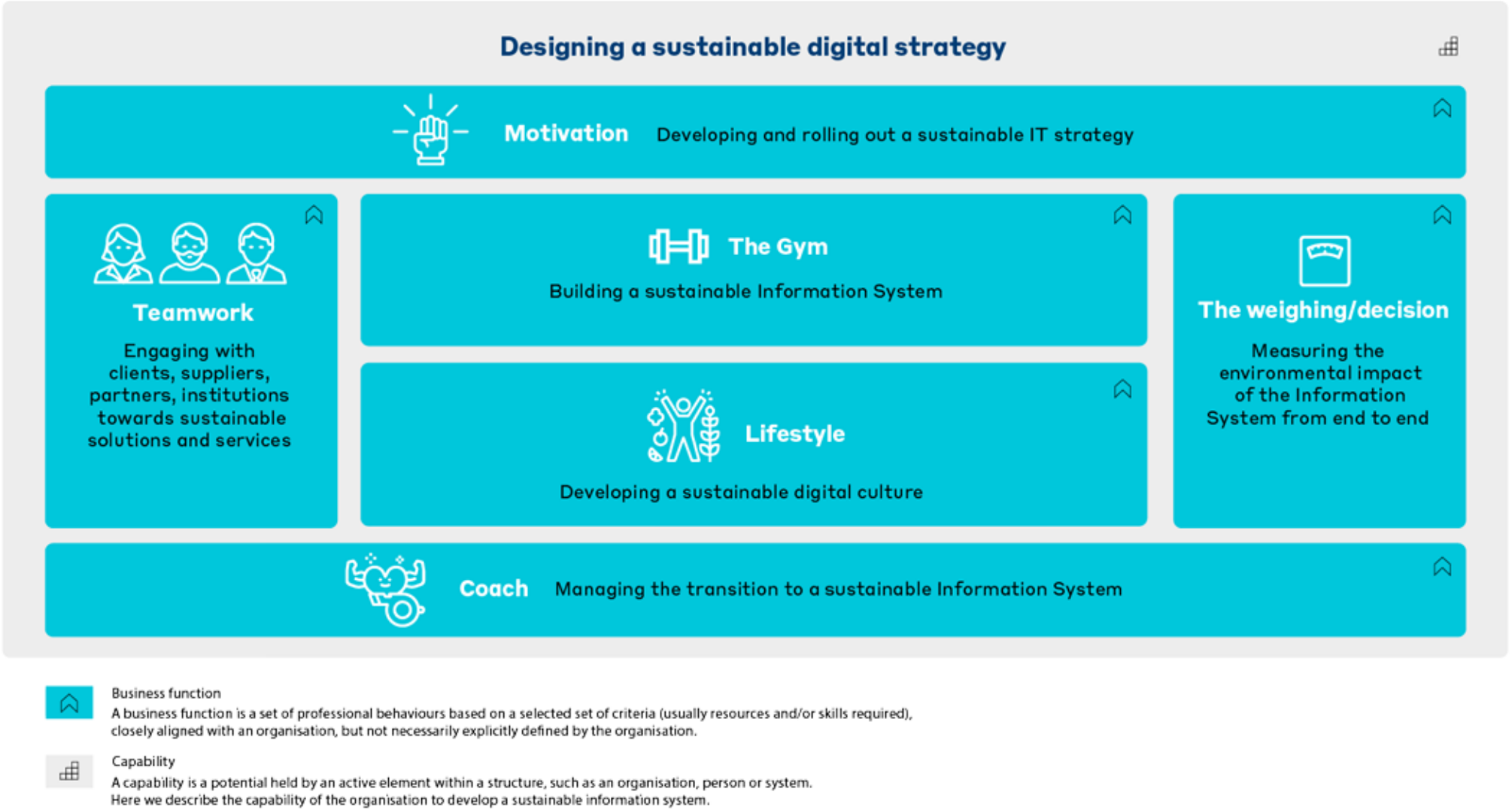
Products	Environmental Impact during manufacturing	Energy efficiency
Screen, Computers, Servers, Phones, Printers	TCO CERTIFIED	ENERGY STAR
Screen, Computers, Servers, Phones	epeat	80 PLUS GOLD
Printer, Phone	BLAUER ENGEL DAS UMWELTZEICHEN	
Computers	EU Ecolabel	

Is my product certified ?

https://tcocertified.com/product-finder/

https://epeat.net/

How to design a sustainable information system ?



Reference model - high level view - "Get the Information System back in Shape". [Source: The Shift Project, production of the working group]

Some steps to evaluate the responsible digital maturity of an organization

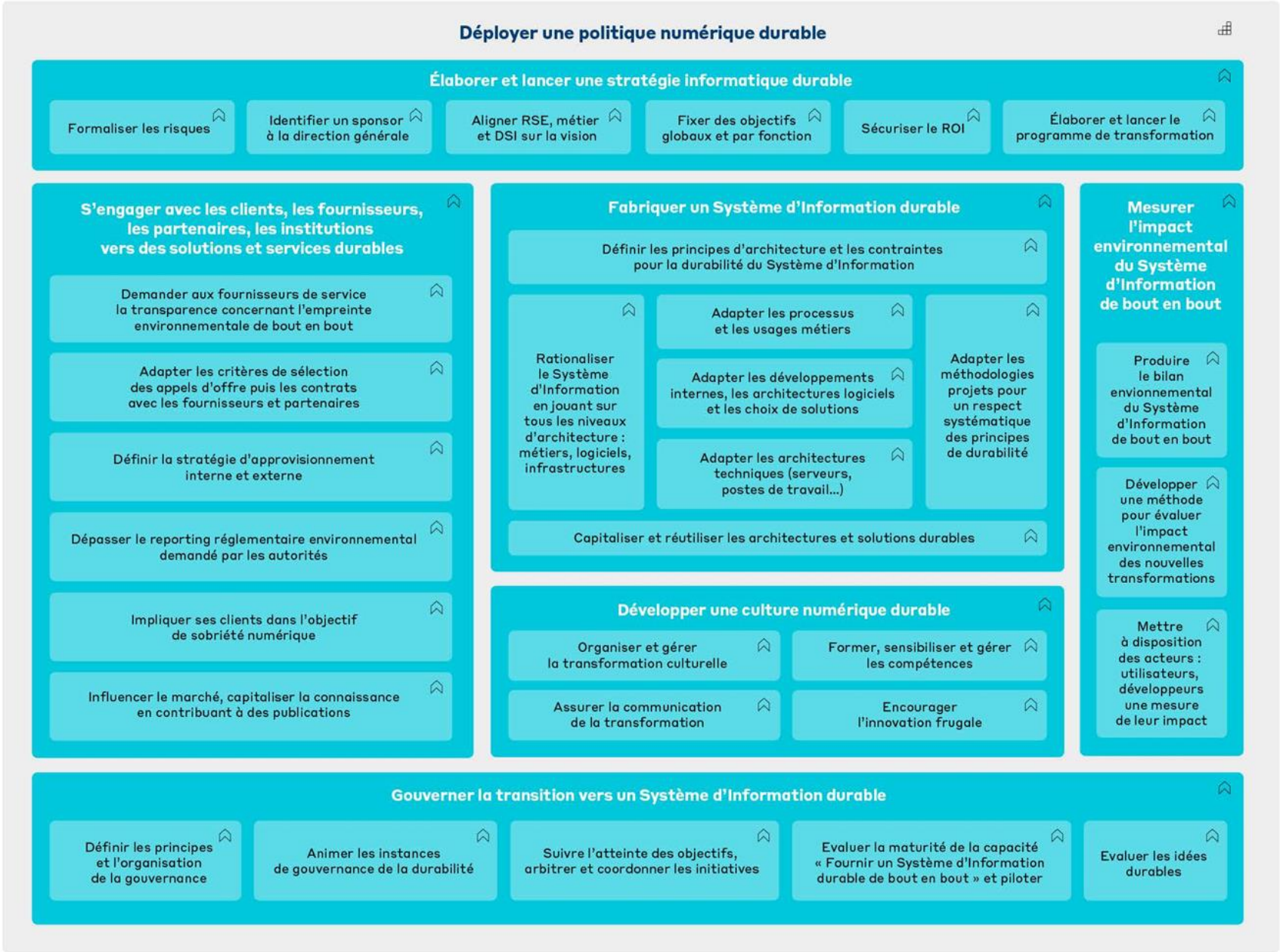
	Responsible Digital Maturity		
	Initial level	**Localised level**	**Managed level**
Priority Level	Very low	Medium	High
Involvement of the CIO and their executive committee	Managers not aware	Executive Committee informed, sometimes driving certain actions Variable level of involvement of managers	Sponsorship from the CIO or CTO Steering committee
Strategy	The CSR strategy may be highly developed, but no RDT strategy has been established	RDT as a strategic focus for the IT department, but it has not yet been formalised An action plan exists but is not always implemented consistently across entities	Cross-functional RDT programme established for a long time Certification processes effective or in progress
Monitoring of the measure	No IT carbon footprint assessment	Annual IT carbon footprint	Ambitious decarbonisation targets, audited and published
Steering of indicators	No KPIs	Quantified targets currently being defined	Precise indicators/KPIs, beyond carbon footprint Integration of cloud and AI into impact measurement
Budget	No dedicated budget	No or little dedicated budget	Budget allocated to RDT Demonstration of ROI
Staff	Individual initiatives, lack of overall coordination.	Awareness-raising underway High degree of heterogeneity in the application of best practices	RDT training requirements for IT, including eco-design Regular awareness-raising among business lines

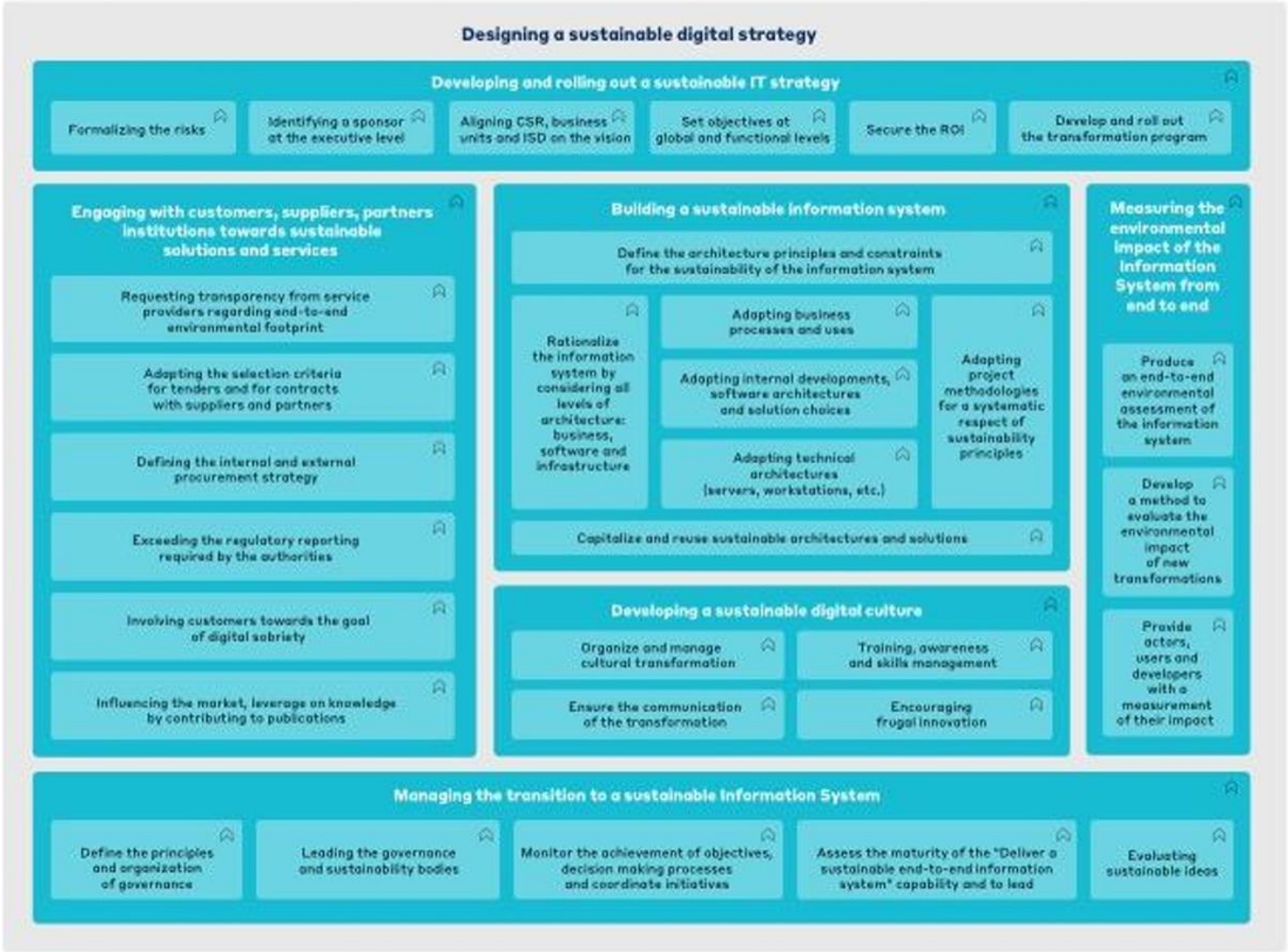
Source : CIGREF



If you are a CTO, which action will you take first to make a more responsible digital ?

Figure 12 - Modèle de référence – vue détaillée - pour une remise en forme du Système d'Information
Source : The Shift Project, production du groupe de travail



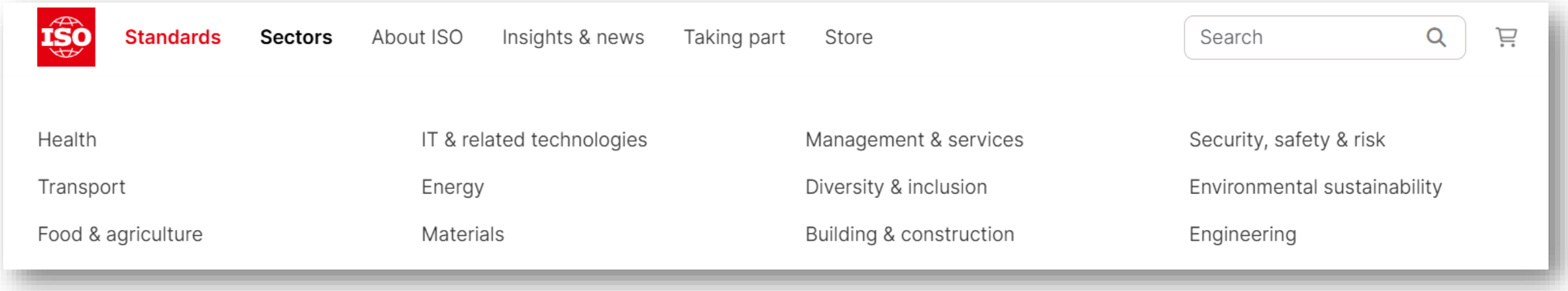


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HEI·ISEN·ISA

STANDARDS

Regulations and Standards

ISO: Global **standards** for trusted goods and services



IT and related technologies

AI | Information security | Coding | Software | IoT | Health IT | Data centres | Audio-visuals | Other

ISO/IEC 30134-9

Information technology — Data centres key performance indicators

Part 9: Water usage effectiveness (WUE)

Published in 2022 — CHF 132

ISO/IEC 30134-8

Information technology — Data centres key performance indicators

Part 8: Carbon usage effectiveness (CUE)

Published in 2022 — CHF 98

ISO/IEC AWI 30134-9

Information technology — Data centres key performance indicators

Part 9: Water usage effectiveness (WUE)

Under development

ISO – IT sector > Datacenter

Abstract

This document specifies water usage effectiveness (WUE) as a key performance indicator (KPI) for quantifying the water consumption of a data centre during the use phase of the data centre life cycle.

WUE is a simple method for reporting the water intensity of the data centre operating. By reporting water consumption, it is possible to present the data centre's resource effectiveness.

This document:

a) defines the WUE of a data centre;

b) introduces WUE measurement categories;

c) describes the relationship of this KPI to a data centre's infrastructure, information technology equipment and information technology operations;

d) defines the measurement, the calculation and the reporting of the parameter; and

e) provides information on the correct interpretation of the WUE.

Life cycle

Previously	Now
Published ISO/IEC 30134-9:2022	→ Under development ISO/IEC AWI 30134-9 Stage: 20.00

SUSTAINABLE DEVELOPMENT GOALS

This standard contributes to the following Sustainable Development Goals

11 13

Regulations and Standards

ISO 50001

Energy management systems — Requirements with guidance for use

Reviewed in 2024 — CHF 155

ISO 50002-1

Energy audits

Part 1: General requirements with guidance for use

Published in 2025 — CHF 177

ISO 50002-3

Energy audits

Part 3: Guidance for conducting an energy audit using ISO 50002-1 in processes

Published in 2025 — CHF 132

ISO 50002-2

Energy audits

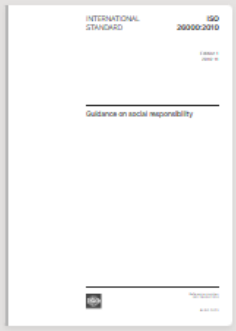
Part 2: Guidance for conducting an energy audit using ISO 50002-1 in buildings

Published in 2025 — CHF 132

ISO 14001

Environmental management systems — Requirements with guidance for use

Published in 2015 — CHF 155



Read sample

ISO 26000:2010

Guidance on social responsibility

Published (Edition 1, 2010)

This publication was last reviewed and confirmed in 2025. Therefore this version remains current.

DATA CENTER & IA



What is a datacenter ?

L'ère des géants de la tech

Entreprises avec les capitalisations boursières les plus élevées au monde, en 2005 et en 2022 (en milliards de dollars) *

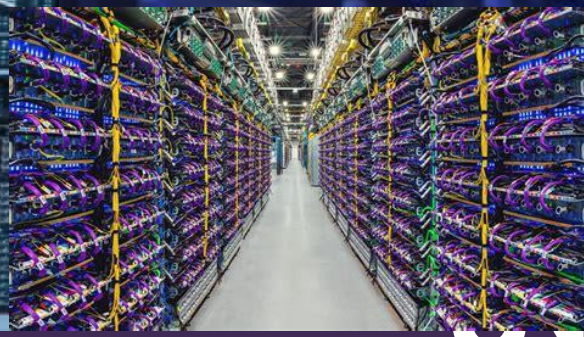
- Tech
- Pétrole / Énergie
- Finance
- Conglomérat
- Distribution physique
- Santé
- Automobile

2005		2022
General Electric	382	
Exxon Mobil	381	
Microsoft	263	2 510 Microsoft
Citigroup	234	
BP	221	
Walmart	212	
Royal Dutch Shell	211	
Johnson & Johnson	200	
		2 986 Apple
		1 927 Alphabet
		1 898 Saudi Aramco
		1 728 Amazon
		1 205 Tesla
		942 Facebook
		751 NVIDIA





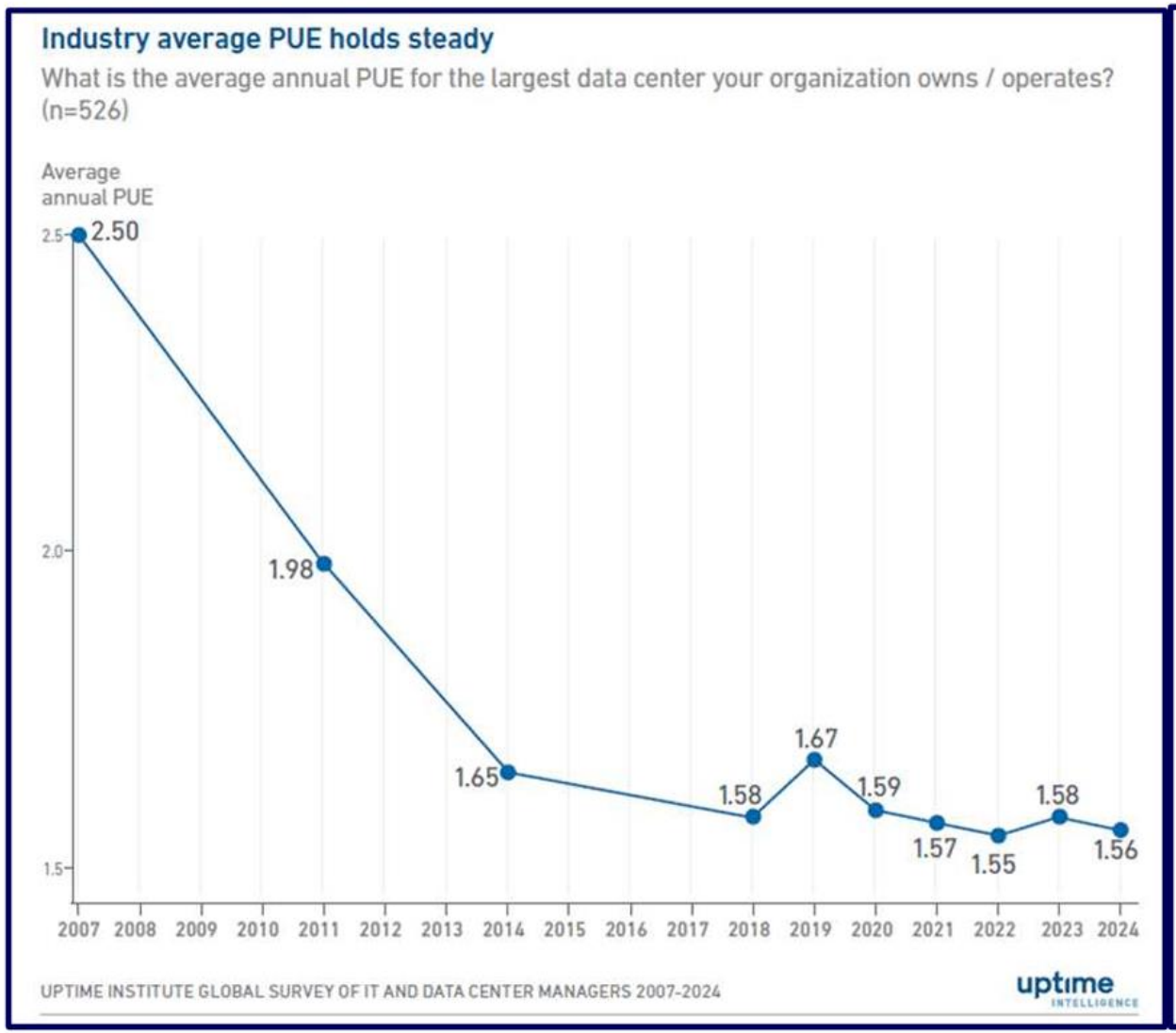




Datacenter

	INFRASTRUCTURE Building and technical environment 	**IT INFRASTRUCTURE** 	**SERVICE** 
Delivered Service	Provision of secure hosting, supplied with high-quality electricity and an environment compatible with computer equipment	Provision of IT equipment hosted in a data center	Provision of computing resources in the cloud or applications
Indicators	PUE (Power Usage Effectiveness) CUE (Carbon Usage Effectiveness) Power Consumption Energy Performance	Environmental impacts: life cycle per hosted server, storage, network equipment, etc.	Environmental impacts life cycle by VM, CPU, RAM, Business act…

PUE = Total facility power / IT equipment energy



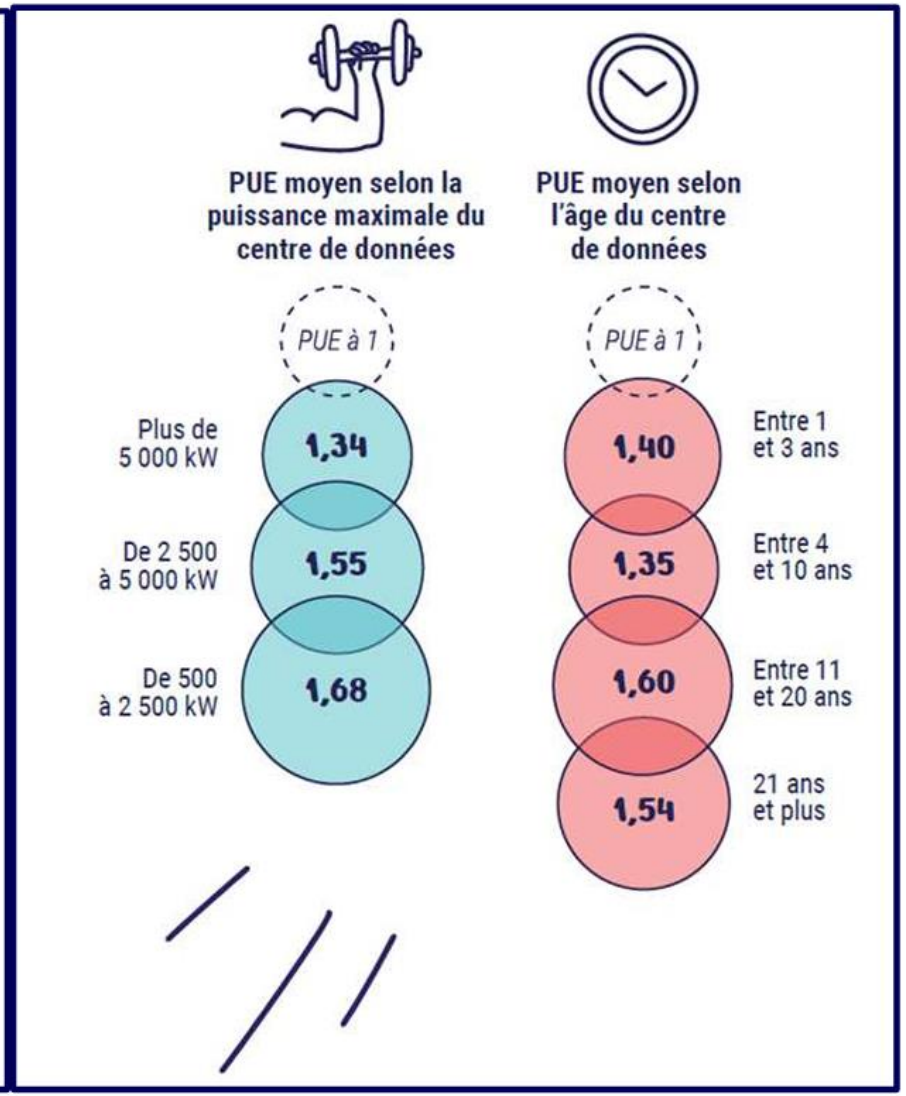


Figure 13 - Evolution du PUE. Sources : (Arcep, 2023; Uptime Intelligence, 2024)

What are Data Center Tiers?

Data center tiers are a classification system developed by the Uptime Institute to describe the expected availability and fault tolerance of a facility.

Each tier builds on the previous one, adding more redundancy and uptime guarantees - but also higher costs and longer construction timelines.

- **Tier I:** Basic infrastructure, limited redundancy
- **Tier II:** Adds redundancy for critical components
- **Tier III:** Concurrent maintainability, higher uptime
- **Tier IV:** Fault-tolerant, maximum uptime and redundancy

What are Data Center Tiers?

	Tier 1	Tier 2	Tier 3	Tier 4
Uptime guarantee	99.671%	99.741%	99.982%	99.995%
Downtime per year	<28.8 hours	<22 hours	<1.6 hours	<26.3 minutes
Component redundancy	None	Partial power and cooling redundancy (partial N+1)	Full N+1	Fault tolerant (2N or 2N+1)

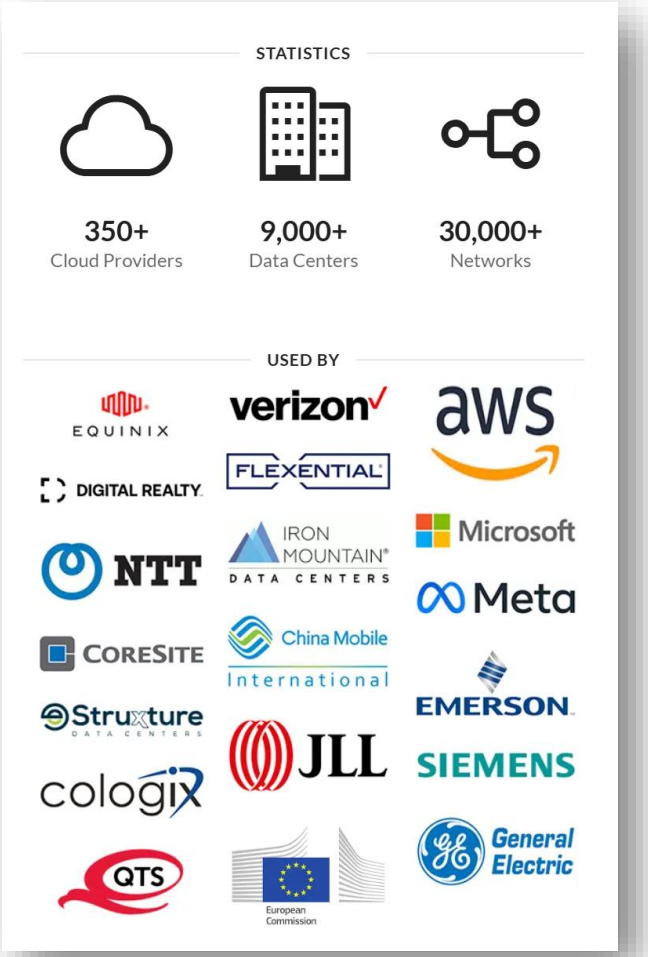
https://mhd-experts.com/2023/05/31/classification-des-centres-de-donnees-en-niveaux-comprendre-la-fiabilite-et-la-disponibilite/

How many datacenters ?

https://www.datacentermap.com/

https://aws.amazon.com/fr/about-aws/global-infrastructure/

https://cloud.google.com/about/locations?hl=fr



Google - Data Center Locations

110 DATA CENTERS

66 OPERATIONAL

38 UPCOMING

16 COUNTRIES

47 MARKETS

And they need electricity…

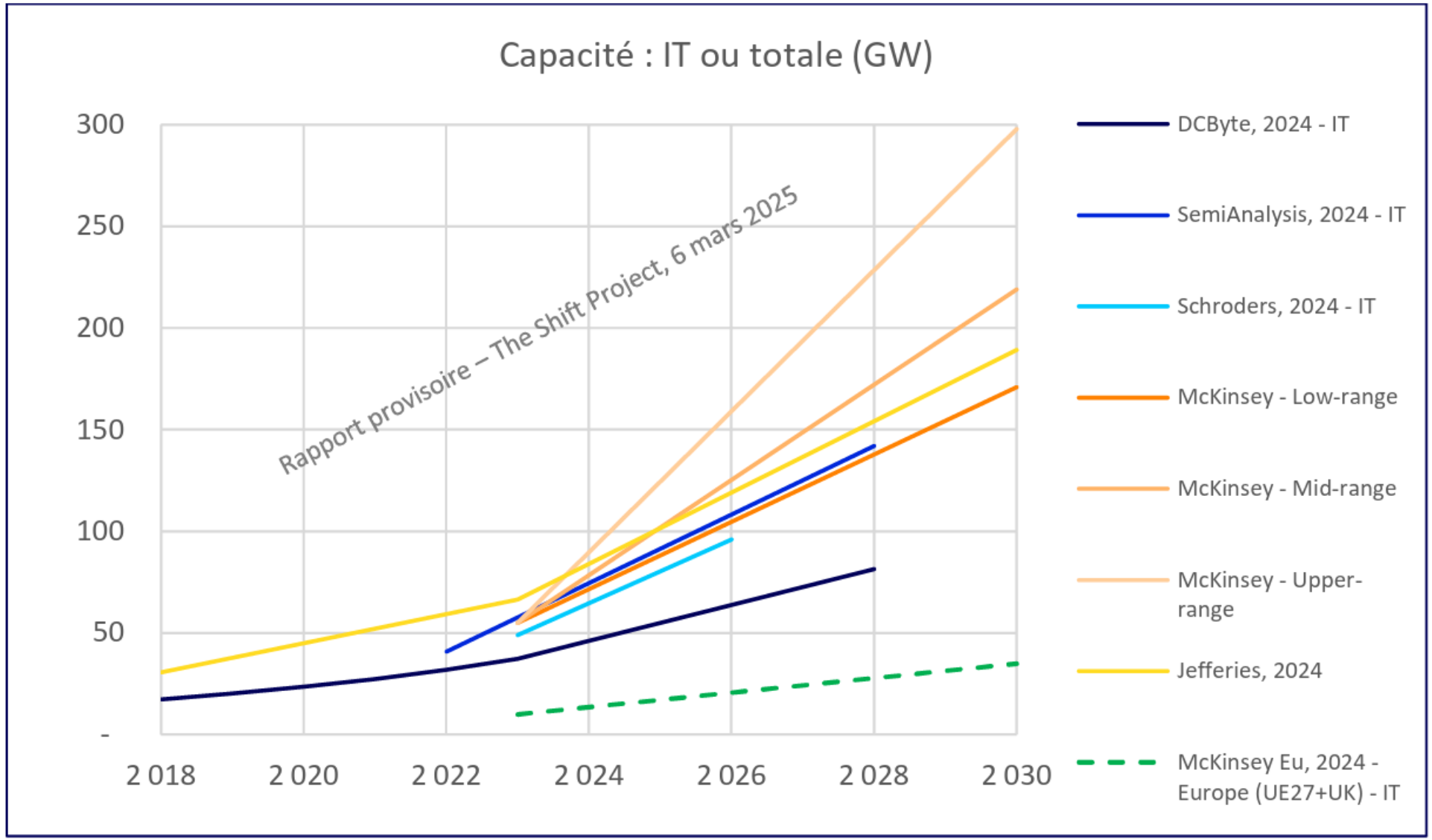


Figure 7 - État des lieux des prévisions de demande de puissance électrique pour l'IT ou pour les centres de données dans les centres de données mondiaux. Sources : (DCByte, 2024; Jefferies, 2024; McKinsey & Company, 2024; McKinsey, 2024; Schroders, 2024; SemiAnalysis et al., 2024)

Sources : https://theshiftproject.org/publications/intelligence-artificielle/

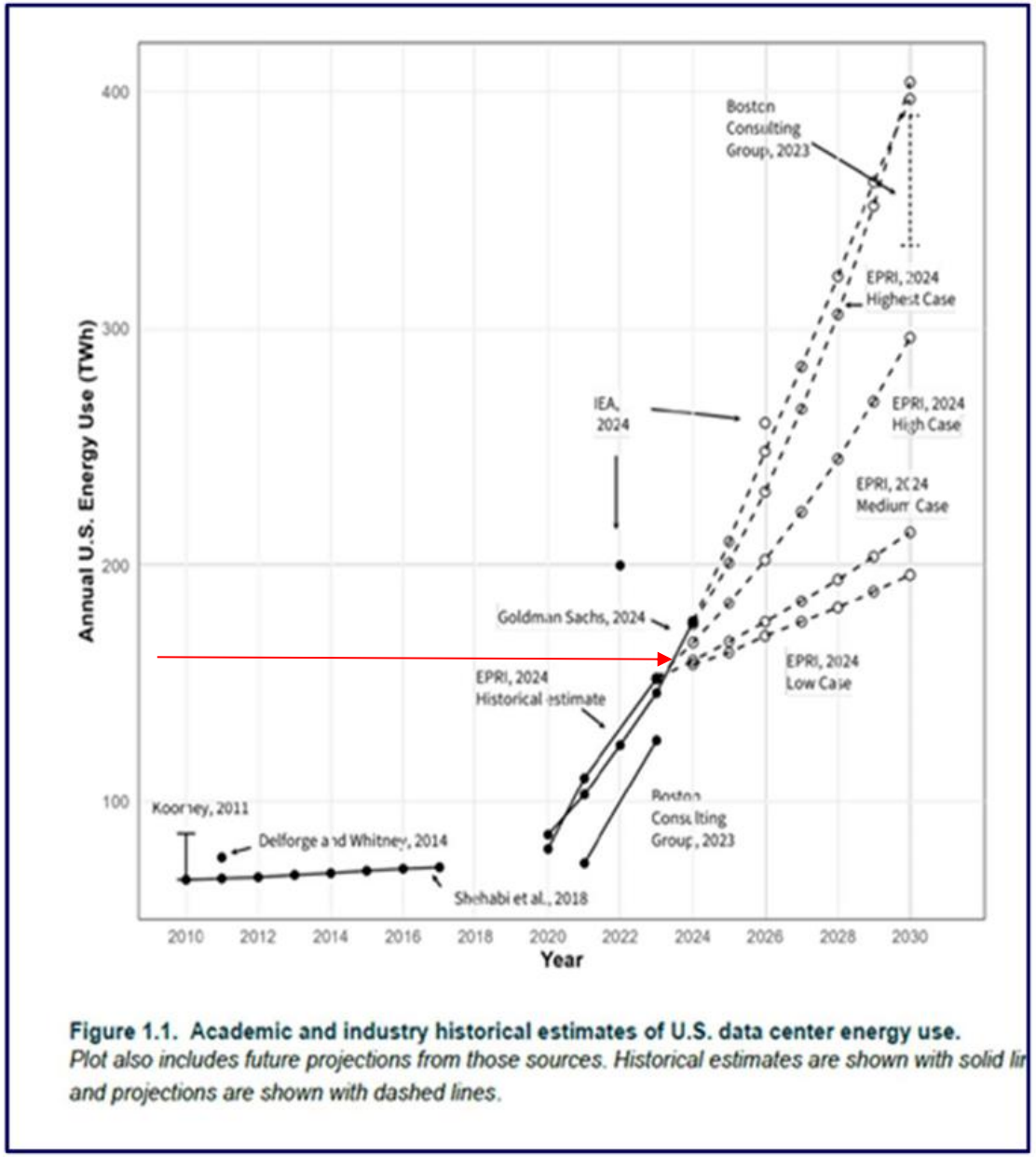


Figure 1.1. Academic and industry historical estimates of U.S. data center energy use.
Plot also includes future projections from those sources. Historical estimates are shown with solid lir and projections are shown with dashed lines.

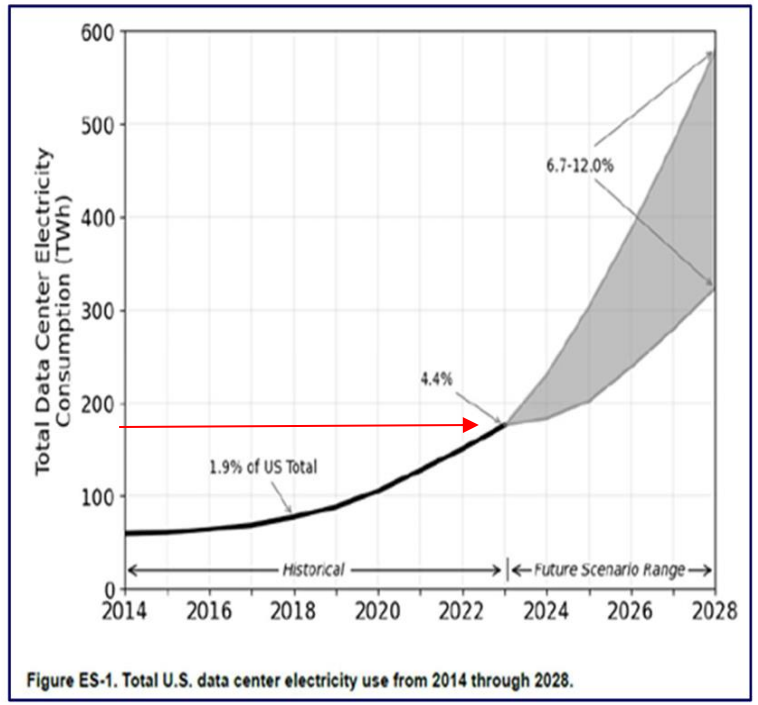
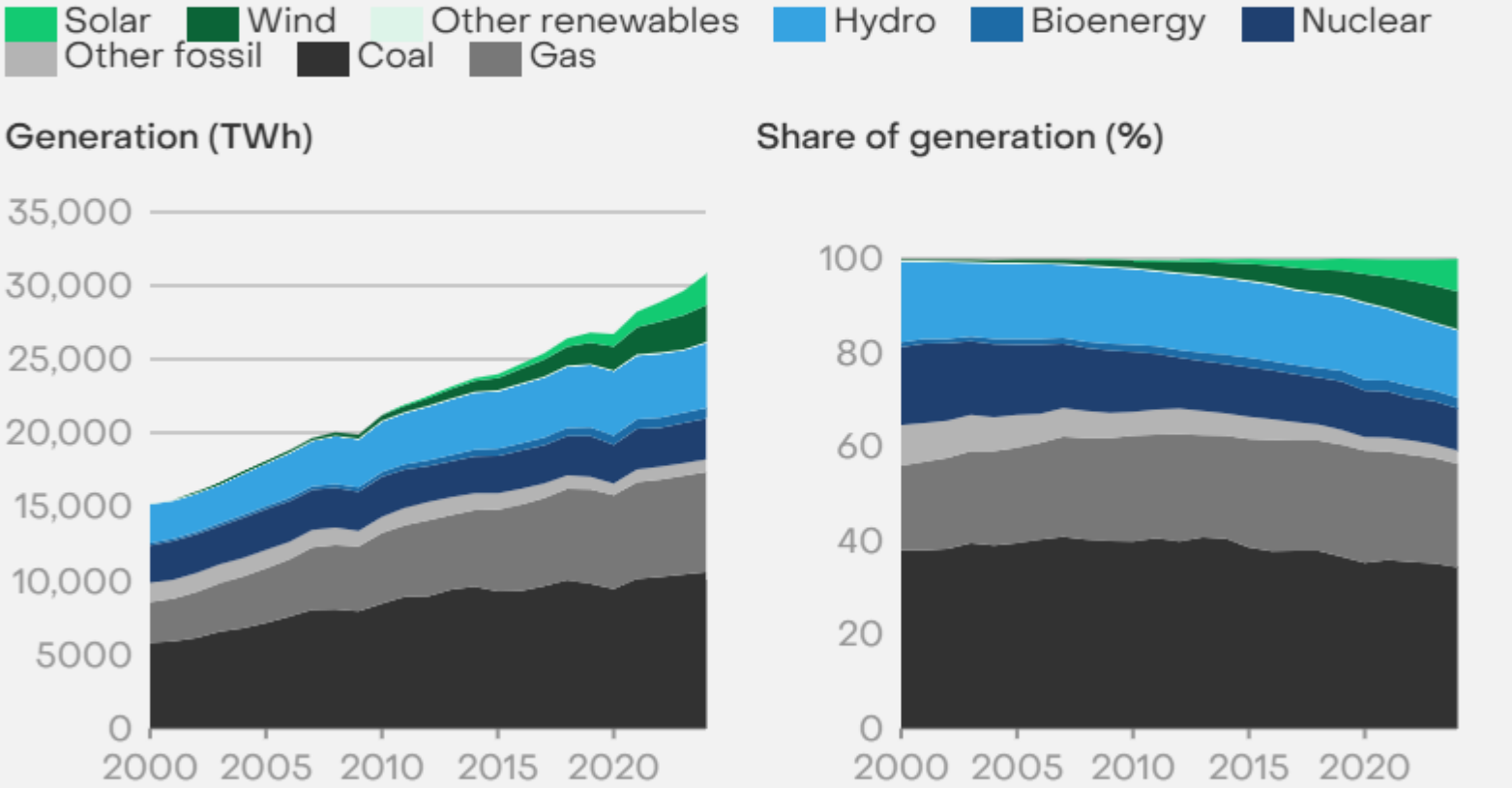


Figure ES-1. Total U.S. data center electricity use from 2014 through 2028.

Figure 5 - Historique de la consommation électrique des centres de données entre 2014 et 2023 aux
Source : LBNL, 2024

Figure 4 - Historique des estimations en consommation électrique des centres de données entre 2020 et 2030 aux Etats-Unis Source : (LBNL et al., 2024)

How the global electricity mix has evolved over time



Source: Yearly electricity data, Ember

EMBER

Investment in DC

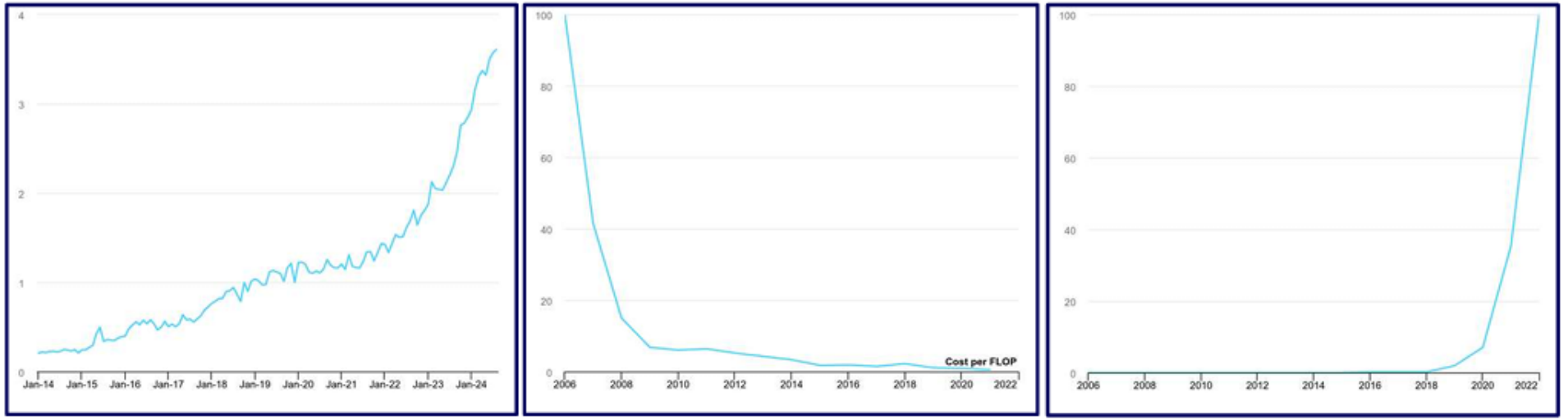


Figure 10 - Investissements dans les centres de données aux Etats-Unis (à gauche)[19], coût des processeurs graphiques (au centre)[20], taille et complexité des modèles d'intelligence artificielle (à droite)[21]. Source : (IEA & IEA Paris, 2024c, 2024b, 2024a)

- **Machine learning** : increase in hardware and algorithmic performance, in model performance and in the computing power required to train "remarkable" machine learning models

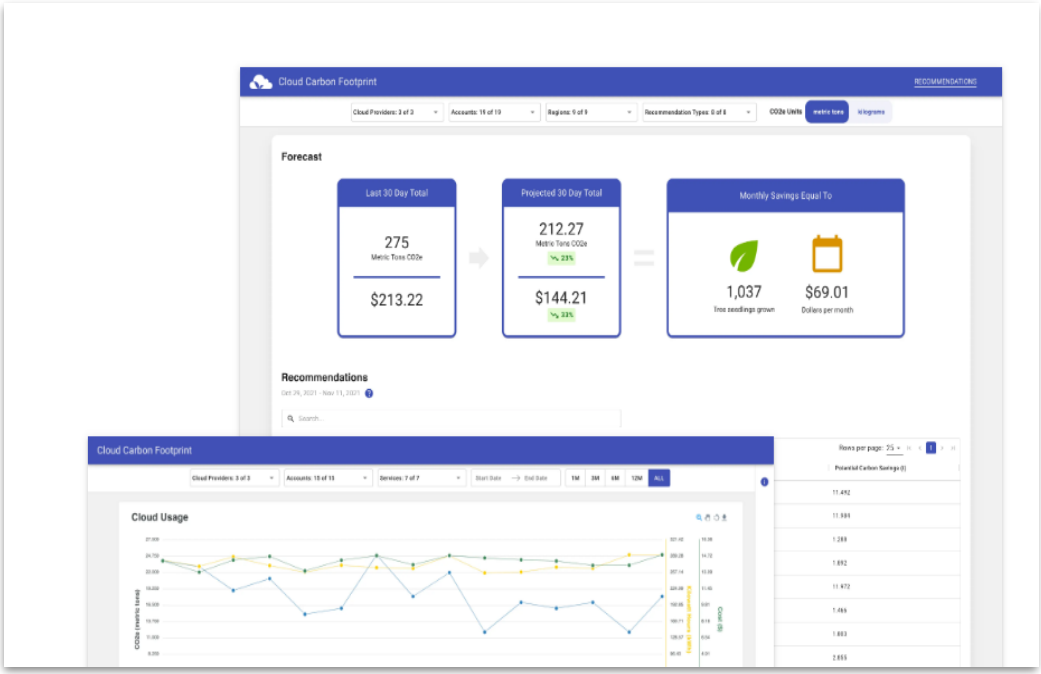
increase in electricity consumption

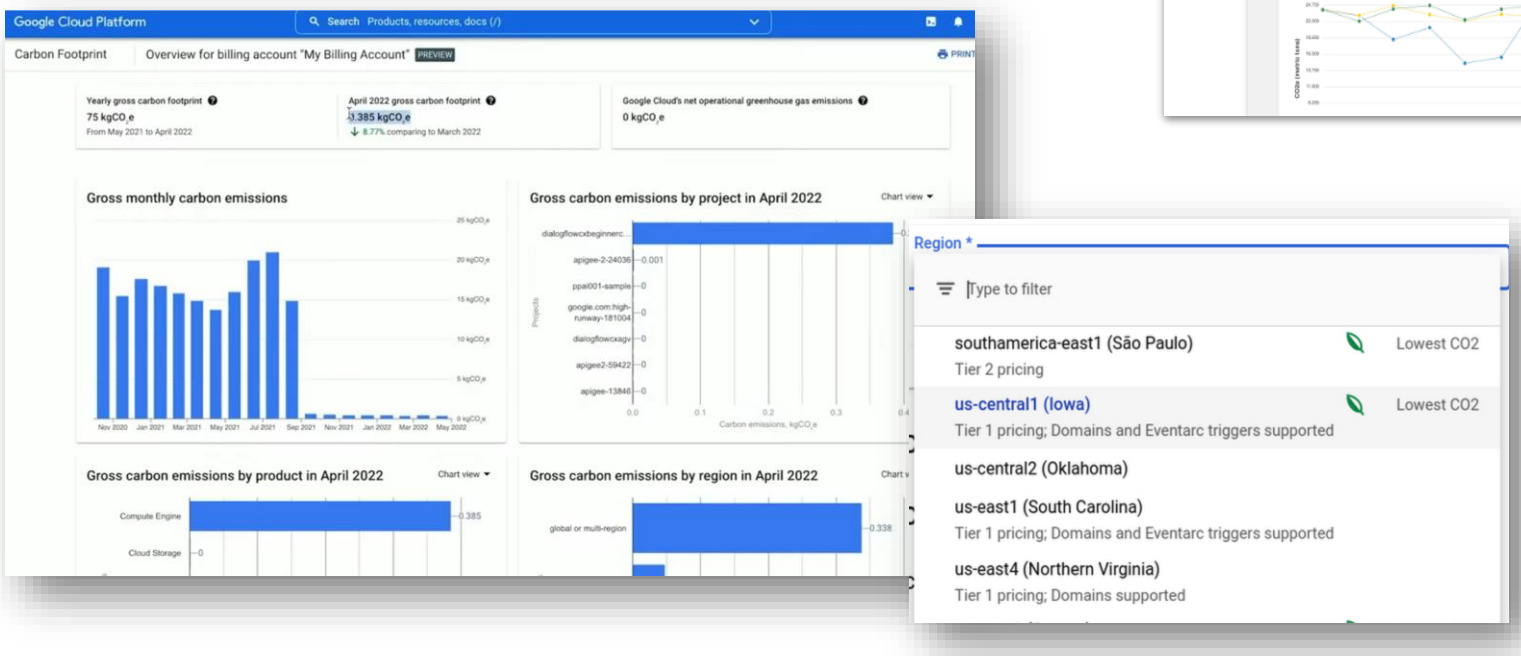
accelerated replacement of less powerful equipment

How to monitor cloud impact ?

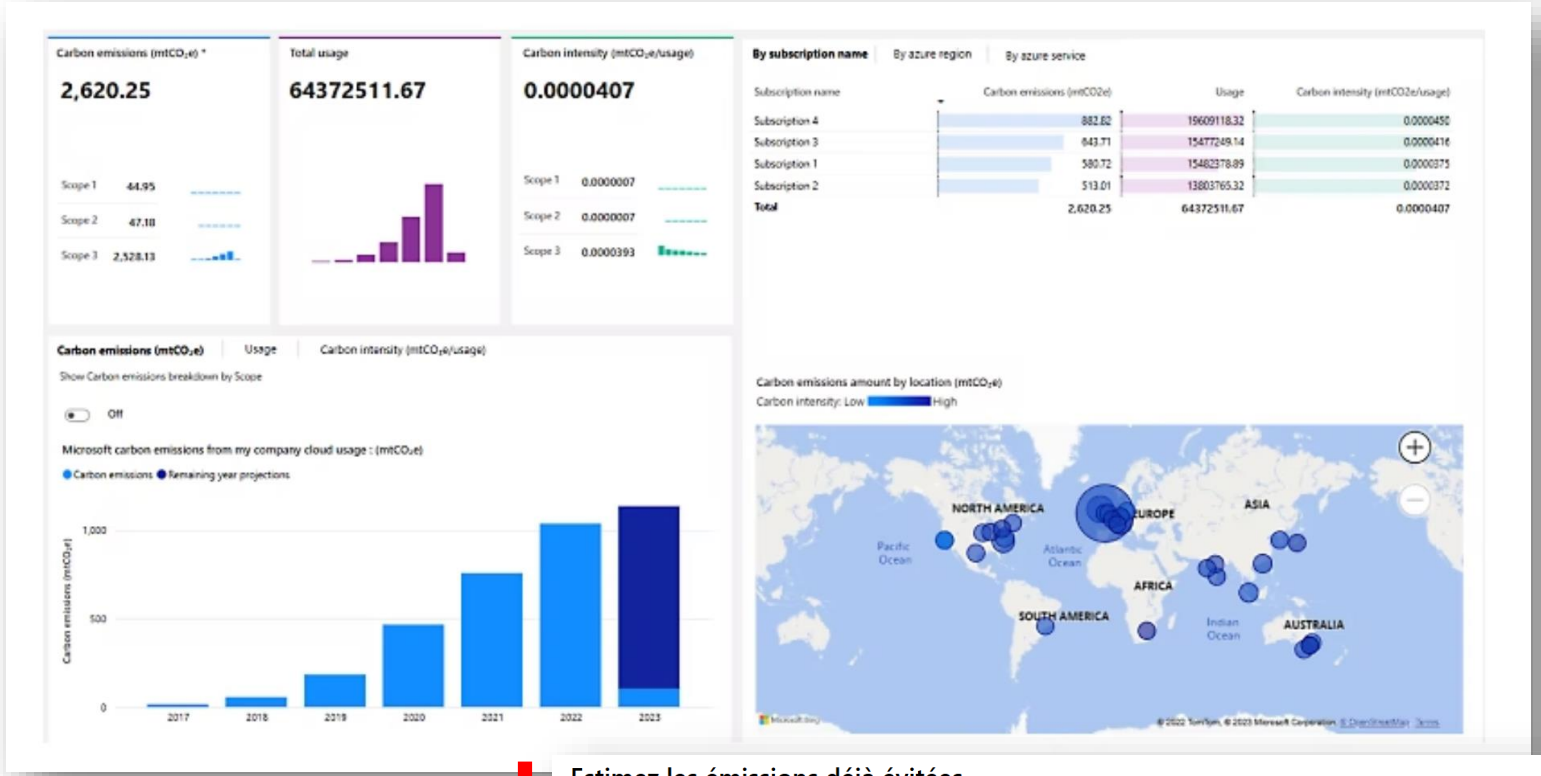
Open-source tool : https://www.cloudcarbonfootprint.org/

Google





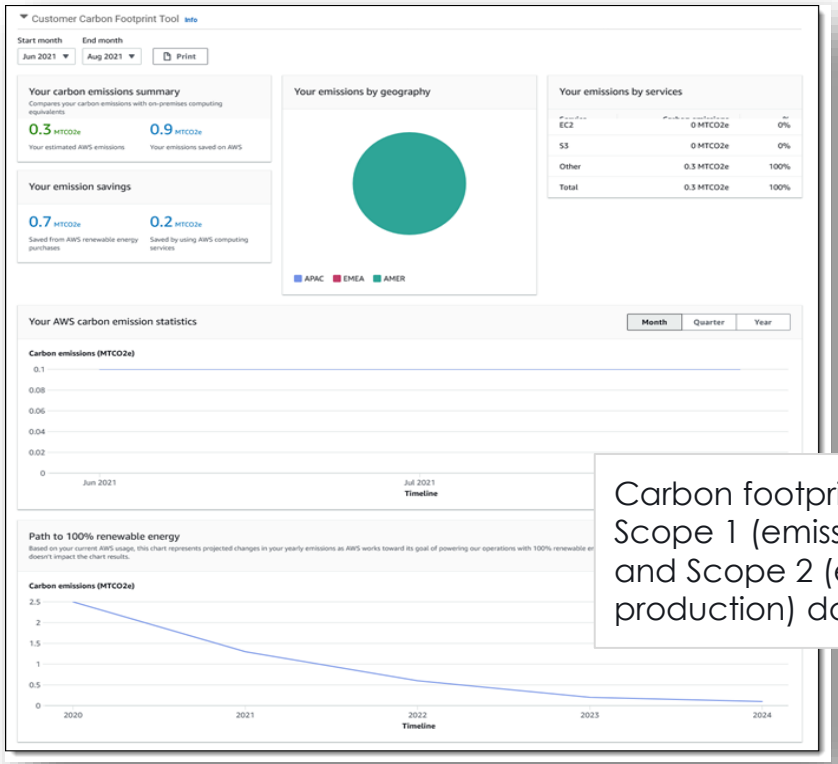
Azure



Estimez les émissions déjà évitées

Calculez les émissions déjà évitées par votre organisation grâce à la migration de vos charges de travail locales vers Microsoft Cloud.

AWS

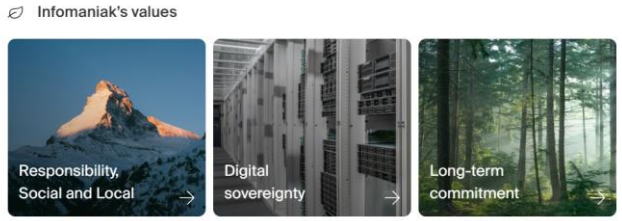


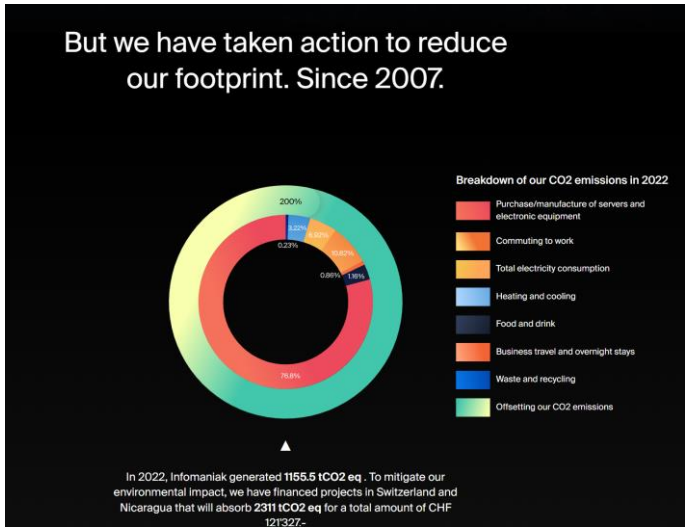
Carbon footprint estimates for AWS include Scope 1 (emissions from direct operations) and Scope 2 (emissions from electricity production) data.

Some example of alternative hosting solutions

infomaniak

https://www.infomaniak.com/en/ecology





HOSTERRA

https://hosterra.eu/en/

Hosterra is a young company that does not yet have its own data centers. We have therefore chosen, for now, to subcontract our data centers to operators who respect very strict specifications. These operators must be French, have 100% of their infrastructure in Europe, propose an agreement of non transfer of data outside the European Union, be energy efficient (with a PUE lower than 1.6) and water efficient (with a WUE lower than 0.3).
Hosterra currently offers services at the following data centers:

Saint-Ouen l'Aumône, Paris, France		Vitry-sur-Seine, Paris, France	
Commissioning	2018	Commissioning	1991
Size	17000 m²	Size	5500 m²
Cooling	Direct and adiabatic freecooling	Cooling	Chilled water system
Operator	Scaleway	Operator	Scaleway
Electrical capacity	2x24 MW	Electrical capacity	2x9 MW
Power supply	100% renewable (hydro + wind)	Power supply	100% renewable (hydro + wind)
PUE class	1.2	PUE class	1.5
Water consumption	<3300 m³/an	Water consumption	<270 m³/an
WUE class	0.2	WUE class	0.02
Monitoring	real-time KPIs	Monitoring	real-time KPIs

VPS (virtual private server)

18 kWh	100%	0.18 kg	1.58 l	+ 16.9°C
ENERGY	RENEWABLE	CO_2	WATER	Δ-TEMP

"green" hosters per country : https://www.thegreenwebfoundation.org/directory/

As a conclusion

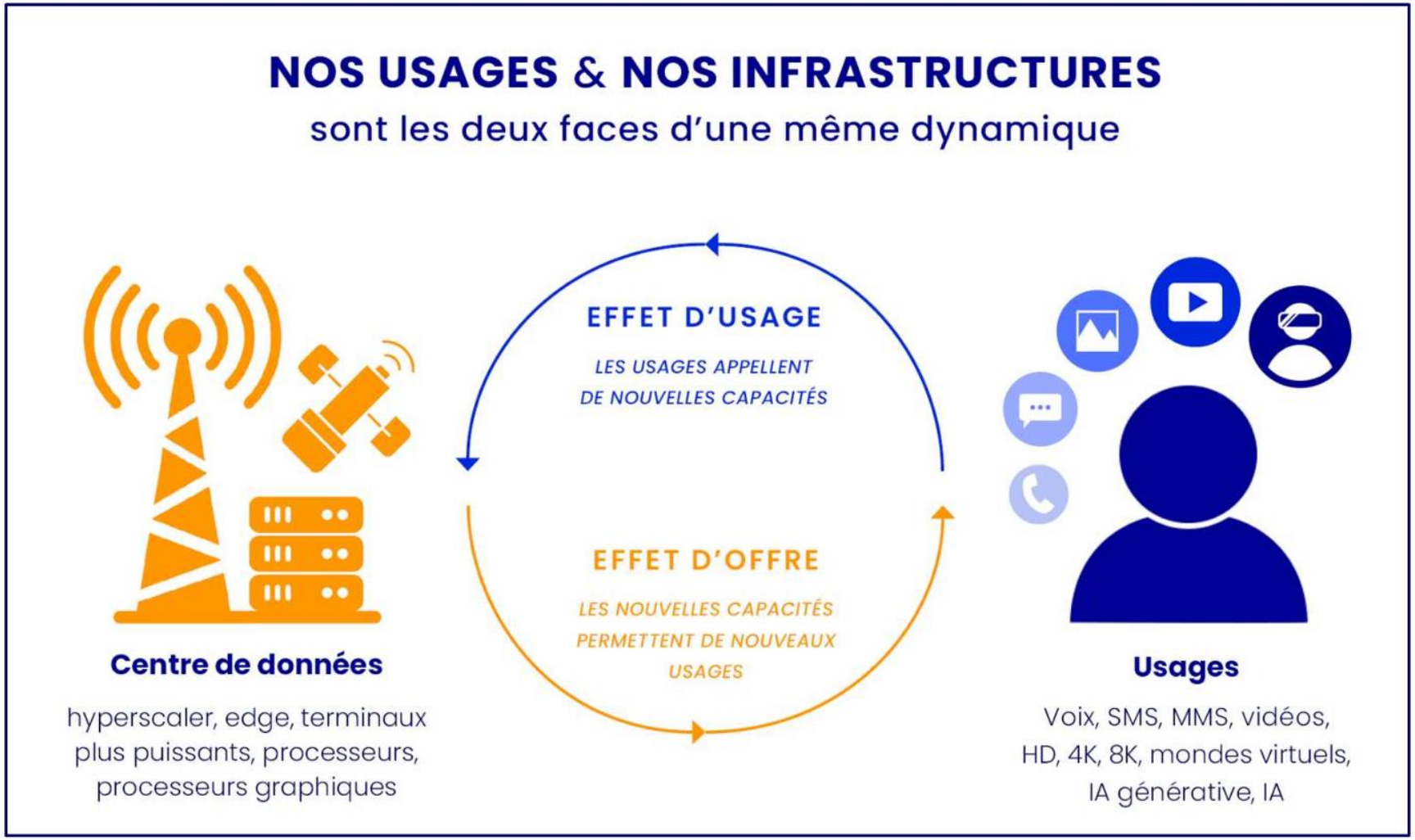


Figure 2 - Nos usages et nos réseaux sont les deux faces d'une même dynamique.
Source : Rapport intermédiaire, The Shift Project, dans le cadre de ce rapport

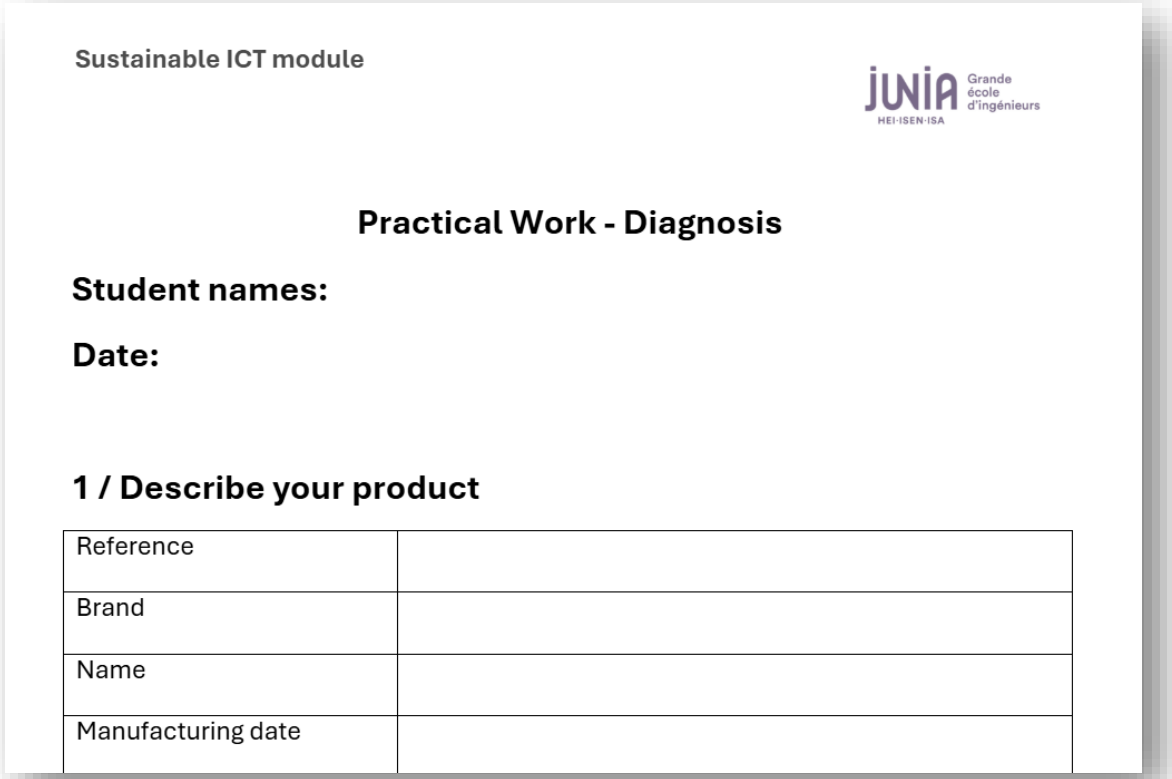
Lesson 2 - As a summary

- Design a sustainable information system
  - Measure (mono / multi criteria)
  - Coach (Engage people and teams)
  - Sustainable sourcing
  - Sustainable software development
- Refer to standards & Ecolabels
- Data and IA accelerate datacenter intensity
- Minimize datacenter impact :
  - PUE closed to 1
  - Renewable energy
  - Keep as long as possible the equipments

Technology is moving fast. Use, costs and culture are challenging to move forward a sustainable IT

PRACTICAL WORK

TP diagnosis



Sustainable ICT module

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Practical Work - Diagnosis

Student names:

Date:

1 / Describe your product

Reference	
Brand	
Name	
Manufacturing date	

TP - Practical Work - Diagnosis.docx

QUESTIONS

JUNIA Grande école d'ingénieurs

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Septembre -Décembre 2025

ISEN
Module
Numérique Responsable